Occupation-Subsector Complementarity on Establishment-Level Growth

Aidan Soares Oliveira

August 2025

Abstract

I discuss the role of subsector–occupation complementarity, which I argue to be a method of quantifying labor specialization, on establishment-level growth using RAIS data for years 1985–1993. Specialization is shown to have a positive effect on establishment size. The relationship is robust under several specifications. The data suggest large establishments can derive more growth-enhancing benefits from specialization than small establishments. The role of specialization on other productivity measures is analyzed, with results differing for tradable vs. nontradable subsectors. Finally, the role of density and agglomeration is explored. There is suggestive evidence that a more specialized workforce contributes to higher establishment growth through localization economies rather than urbanization economies, though more research is needed.

Acknowledgements

This thesis was a culmination of shifting horizons, changes in directions, tedious trudges, puzzlement, but the experience, in all its multitudes, is one I wouldn't trade for anything; it was precisely because, rather than despite, the steepness of the intellectual climb that I was able to get a glimpse, however fleeting, of intellectual horizons so hypnotizing. This thesis was a culmination of attempts at seeking, of steadfastly exploring, a question into which I have sought inquiry for many years now. And, I would like to continue inquiring into for many years to come: the question of what makes our cities so fascinating, what produces the agglomerations of workers, firms, and talent that captivates so many of us, what kind of economic forces are these, without which a city would simply not be a city.

I admit that my thesis, while remaining faithful to that mission, has left more questions open then it has answered. In producing microfoundations of greater granularity, less was uncovered of those very urban forces whose inquiry motivated my search to begin with. Such is the case for all productive research. Good questions lead to better questions. Nothing is final. Research, in its broadest manifestation, is a lifelong journey. And, that journey is not one worth making, just as I think no research is worth doing, no questions worth asking, no meaningful answers worth giving, without a community of other people as interested in these questions as I am.

It is with great pleasure, that I acknowledge everyone falling in that category, those who make up that community, and let each of you know, that you are not just actors on this journey I am on. You are not just set pieces, background forces. That much is obvious. But, what I will say next, perhaps less so. You are more than just mentors, advisors, and professors together on this journey with me, although, it is true, you were all those things. You form the very essence of the journey itself. Simply put, there would be no journey, as I come to understand that word, without each of you present in it.

In my academic endeavors, I have received ample help and support, and, consequently, I have many to acknowledge. Included here are all of those who, in early phases of this project, back when I was just brainstorming and had little idea where the project would go, took interest in what I was doing and provided valued advice. This includes Minchul Shin, Simon Freyaldenhoven, Enghin Atalay, Jeffrey Brinkman, Gerald A. Carlino, Satyajit Chatterjee, Jeffrey Lin, Kyle Mangum, and Bryan Stuart, all of whom showed genuine interest in helping me formulate my research question and gave helpful feedback on some of my early ideas back when I was interning at the Philly Fed. I also want to thank Gilles Duranton who was nice enough to discuss my proposal over a zoom call despite us not having met before that point.

I want to thank all of the Tufts faculty for their kindness, generosity, and desire to help. I want to thank Adam Storeygard for listening to an early version of my proposal and giving me excellent feedback. I want to thank Federico Esposito for being a bulwark of support back when I had wished to utilize a trade shock to identify the effect of my variable of interest. Although that endeavor did not come into fruition, I remember each of the ideas we formulated, each of the proposals we discussed, and, on a future date, I wish to implement them once I get access to more detailed data.

I want to thank Enrico Spolaore and Alan Shapiro for the countless talks they gave during the Thesis Seminar. The advice given was helpful, not only for my own thesis research, but for all research I hope to do in the future. I want to thank Marcelo Bianconi for taking great interest in my success and giving me excellent feedback throughout this process.

Most importantly, I want to thank Jeffrey Zabel, both for his role as my thesis advisor but also

for everything that came before. He was an excellent professor, had captivating lectures (a rarity in academia these days), was thoughtful, diligent, and, there was never a time I felt he wasn't eager to cultivate my intellectual growth. He had insightful probing question every step of the way as I was doing my thesis research. I am deeply grateful for how much he valued my intellectual curiosity and his willingness to lend practical support to each of my academic endeavors.

Contents

Acknowledgements **i**

1 Introduction **1**

2 Review of Literature **2**
2.1 Establishment Specialization 2
2.2 Agglomeration Economies 3
2.3 New Work 4

3 Subsector-Occupation Complementarity **5**
3.1 Constructing the Subsector-Occupation Complementarity Score 6
3.2 Modeling Revealed Comparative Advantage 7

4 Data **20**
4.1 Data Sources and Construction 20
4.2 Description of RAIS 20
4.3 Data Cleaning and Matching Procedure 20

5 Main Hypothesis and Empirical Strategy **21**
5.1 Empirical Framework 21
5.2 Summary Statistics and Stylized Facts 22
5.3 Motivation 27

6 Results **28**
6.1 Baseline Findings 28
6.2 Relationship between Specialization and Establishment Size 34
6.3 Comparisons across Productivity Measures 35

7 Conclusion **44**

Appendix **1**

A Model **1**
A.1 Labor Market Structure and Matching Process 1
A.2 4
A.3 5
A.4 8
A.5 10
A.6 Relationship to Jara-Figueroa et al. (2018) 13

1 Introduction

This paper investigates the role of labor market specialization in shaping establishment-level productivity, measured through average wages, labor force size, and total wages. To quantify labor market specialization, I adapt a measure from the international trade literature—the Balassa index, or revealed comparative advantage (RCA) index—for use in the context of occupations and subsectors within the labor market. Whereas Balassa (1965) originally developed the index to capture revealed comparative advantage across countries in international trade, I repurpose the concept here to measure the comparative advantage that occupations exhibit within particular economic subsectors.

The motivation for this inquiry is twofold. First, I aim to better understand the role of establishment specialization as a response to technological shocks, the application of knowledge, and competitive pressures. A growing body of research has deepened our understanding of firm organization and internal labor market practices, particularly as they relate to specialization. For instance, Backus (2020) demonstrates that competitive pressure induces firm reorganization, with successful firms exhibiting greater specialization. Bessen (2015) shows that firm specialization can emerge as a response to technological innovations that lower barriers to entry in certain occupations. Relatedly, Jara-Figueroa et al. (2018) study the effects of "industry knowledge" and "occupational knowledge"—which workers carry when switching jobs—on the growth of pioneer firms. Lin (2011) explores how technological shocks create new work by spurring the reorganization of labor and production.

The second goal is to examine the interaction between establishment specialization and agglomeration economies. In studying the emergence of new work, Lin (2011) grounds his understanding of agglomeration in the tradition of Jacobs (1969), emphasizing urbanization economies. Agglomeration externalities can generally be categorized into two types: urbanization effects and localization effects. Urbanization effects (Jacobs externalities) refer to benefits firms receive from operating in cities with a diverse industrial base, while localization effects (Marshall-Arrow-Romer externalities) pertain to the benefits firms accrue from proximity to other firms in the same industry.

A large literature examines the relative importance of these forces. It is well-recognized that urbanization and localization are not mutually exclusive, and the benefits a firm derives may depend on its stage in the life cycle. Duranton & Puga (2001) model firm location decisions across French employment areas between 1993 and 1996, proposing that firms initially locate in dense, diverse cities to learn about their production processes and later relocate to smaller, more specialized cities to access a more tailored labor force at lower cost.

To shed light on the roles of specialization and agglomeration economies in establishment-level productivity, the paper proceeds as follows. Section 2 reviews the relevant literature.Section 3 introduces the subsector-occupation complementarity measure employed in the analysis. Section 4 describes the data collection and cleaning process. Section 5 presents the main hypothesis, provides motivating evidence, and links the findings to agglomeration economies. Section 6 presents the empirical results. Section 7 concludes.

2 Review of Literature

2.1 Establishment Specialization

I define a "specialized establishment" as one that hires a more focused set of occupational groups. For a concrete example, Eeckhout (2021) describes the case of Nancy, a receptionist working for Eligio, a company that provides receptionists to other firms. Historically, such roles would have been performed in-house. However, domestic outsourcing has become increasingly prevalent. Goldschmidt & Schmieder (2017) document a dramatic rise in domestic outsourcing in Germany since the early 1990s, showing that firms outsource peripheral functions to cut labor costs, saving more on wages than they spend hiring outsourcing firms.

While part of these cost savings may reflect reductions in employee benefits, outsourcing may also enhance managerial efficiency by allowing firms to focus on a narrower set of occupational groups. As economies grow more complex, occupations become increasingly specialized, requiring more refined management practices. Firms with management structures tailored to a specific occupational composition may enjoy a productivity advantage.

Recent work highlights the importance of managerial quality in labor allocation. Minni (2024) finds that doubling the share of high-quality managers increases output per worker by 61% at the establishment level. Similarly, Metcalfe et al. (2023) find that agglomeration economies can manifest within firms, with effective management playing a central role. Duranton & Puga (2005) document a broader shift in city specialization patterns, from sectoral specialization to functional specialization, with larger cities increasingly specializing in management and smaller cities in production.

Thus, managerial quality is likely a key driver of establishment-level specialization. While the data used in this paper do not allow a direct investigation of management effects, it is important to recognize that establishment specialization likely reflects substantial heterogeneity in managerial practices.

Measuring specialization requires condensing complex occupational data into a tractable index. One approach is to use the Herfindahl-Hirschman Index (HHI) to measure the concentration of an occupation across industries. However, HHI has a key limitation: it makes no distinction between employees who are, and those who are not, employed in their area of comparative advantage. It is merely a measure of how broadly, or narrowly, an occupation is employed across sectors. For instance, consider the hypothetical of the computer scientist occupation being narrowly employed as determined by the HHI. A computer scientist working outside the tech sector would contribute equally to that HHI-score as one working within it, in so far as a computer scientist, in aggregate, mainly works in a narrow range of sectors; this happens even though only the latter employment is aligned with their comparative advantage.

One method of resolving the dilemma above is to use the measure devised by Balassa (1965) called revealed comparative advantage (or the Balassa index). The measure was designed to be an empirical tool used to approximate the comparative advantage countries have in producing certain products. The index asserts that country i has a comparative advantage in producing product k, compared to the world at large, if the following holds:

$$\frac{(E_{i,k}/E_i)}{(E_k/E)} > 1 \tag{1}$$

where $E_{i,k}$ represents the exports of product i coming from country k, E_i represents the total

exports coming from country i, E_k represents the total exports of product k across all countries, and E represents the total exports in the world at large.

The revealed comparative advantage (RCA) can be adapted to estimate the comparative advantage that occupations have in different subsectors. Jara-Figueroa et al. (2018) employ a similar measure as a robustness check in their study of firm growth and worker knowledge flows. I will use the same index in this paper to measure the comparative advantage that occupations have in certain subsectors of the economy.

It is important to note that while Jara-Figueroa et al. (2018) use the RCA index to measure industry relatedness, I use it here to directly measure establishment-level occupational specialization. Section 3 provides a formal justification for this adaptation, drawing on theoretical frameworks from Eaton & Kortum (2002) and French (2017).

Of course, the RCA measure has limitations. It abstracts from the underlying supply and demand forces that shape occupational distributions across subsectors. Concentration may reflect occupational demand across subsectors, not just comparative advantage. Nevertheless, under plausible conditions (discussed in Subsection 3.2), the RCA measure provides a meaningful proxy for occupational-subsector complementarity.

2.2 Agglomeration Economies

A substantial body of research connects agglomeration economies to firm- and establishment-level productivity outcomes. The consensus is clear: denser areas confer a productivity premium. Faberman & Freedman (2016) survey the evidence, finding that most estimates of the elasticity of earnings with respect to urban density lie between 0.03 and 0.10.

Several mechanisms have been proposed to explain this premium. Syverson (2004) shows that in more competitive cities, the productivity distribution for the ready-mix concrete industry exhibits a higher mean and is truncated from below. Syverson argues that this pattern reflects stronger firm selection: more competitive areas drive out less productive firms, thus raising average productivity.

Gaubert (2018) highlights another channel: firm sorting. She finds that high-density, competitive areas attract more productive firms, suggesting that part of the observed productivity premium stems from the endogenous location decisions of firms, not just competition-induced selection.

However, other work complicates the simple selection story. Combes et al. (2012) use comprehensive French firm data across sectors and find that agglomeration economies—not selection—explain most of the productivity differences between cities. Their framework, building on Metcalfe et al. (2023) and Del Gatto et al. (2011), allows for both selection and agglomeration but finds that the direct productivity-enhancing effects of agglomeration dominate. Interestingly, they find that firms higher up in the productivity distribution benefit more from density than those at the lower end.

Accetturo et al. (2018) similarly find that while firm selection matters, agglomeration effects are primary. Using a model that incorporates varying spatial ranges of agglomeration and heterogeneous market access across regions, they estimate that only between one-fifth and one-third of the total factor productivity (TFP) differential between agglomerated and non-agglomerated areas is attributable to selection.

Particularly relevant for this paper is the work of Backus (2020), who revisits the relationship between competition and firm productivity in the ready-mix concrete industry. Backus employs novel empirical strategies to separate "treatment effects" (direct effects of competition on productivity)

from "selection effects" (the filtering out of less productive firms). He finds that competition exerts a direct positive effect on firm-level productivity that is not explained by selection. Instead, he suggests that competition drives firms to undertake internal reorganization, renegotiate contracts, and increase managerial effort.

Although Backus does not precisely identify the mechanisms of reorganization, he hypothesizes that greater specialization and enhanced managerial inputs are central. As he notes, "the data here suggest greater specialization, greater managerial inputs, and are inconsistent with a story based on capacity utilization." His work strongly suggests that firm-level internal changes—rather than merely market-level selection—are an important source of the productivity benefits associated with agglomeration and competition.

These insights have important implications. The establishment-level productivity gains from specialization may be intimately linked to broader city-level specialization patterns. Establishments that successfully specialize their workforce may be better equipped to identify and prioritize the occupational groups most critical to their production process. Echoing Metcalfe et al. (2023), agglomeration economies are often visible not only across firms but also within firms, particularly in service-sector establishments. As Eckert et al. (2020) emphasize, cities exhibit a comparative advantage in skilled services, underscoring the relevance of these patterns.

2.3 New Work

In addition to facilitating specialization and managerial optimization, urbanization may foster the creation of "new work"—new occupations and organizational structures that emerge in response to technological change. Lin (2011) shows that technological shocks lead to the creation of new job titles and new forms of work organization. Although the data used in this paper do not allow for the direct identification of new occupations or tasks, the process of specialization studied here may reflect these deeper dynamics.

Many modern occupations represent refinements of older, broader categories. For example, "AI researcher" and "data scientist" emerged from the more general category of "computer scientist." Firms that hire a highly specialized set of workers—such as exclusively data scientists or AI specialists—may be positioning themselves to create or exploit new technological niches. In many cases, the specialized tasks that define new forms of work precede formal recognition through new occupational titles.

Lin (2011) emphasizes that urbanization economies—externalities arising from the diversity of industries and knowledge bases in dense cities—are particularly important in fostering new work. Nonetheless, the broader literature documents the importance of both urbanization and localization economies, especially within manufacturing and tradable sectors. Glaeser et al. (1992) find evidence that industrial diversity benefits firms in large cities, consistent with Jacobs externalities. Conversely, Henderson (1986) and Duranton & Overman (2005) find strong evidence of localization economies, particularly in manufacturing industries, even at very fine spatial scales.

Rosenthal & Strange (2003) reinforce this finding by examining six industries (software, food products, apparel, printing and publishing, fabricated metal, and machinery) and finding that localization economies more strongly predict the arrival and employment of new plants than urbanization economies, particularly in tradable industries.

However, as Li & Neffke (2024) and Jara-Figueroa et al. (2018) argue, urbanization and localization economies should not be viewed as distinct forces. Instead, they can be understood

through the lens of relatedness—the degree to which industries share a common knowledge base. Workers benefit from knowledge exchange, but the degree of benefit depends on the similarity of their occupational backgrounds.

Labor flow measures, which track worker movements between industries, provide a powerful way to quantify relatedness. Neffke & Henning (2013) show that relatedness measures based on labor flows predict firm diversification patterns better than standard industry classifications. These insights allow for a more nuanced understanding of agglomeration: industries oriented toward innovation may benefit more from diversity and moderate relatedness, whereas industries focused on cost reduction (rather than product innovation)[1] may benefit more from highly specialized labor pools.

Duranton & Puga (2001) framework, emphasizing firms' location choices as they evolve, helps explain these patterns. In the context of this paper, the subsector-occupation complementarity index—and the theoretical model used to justify its use—provides a framework for conceptualizing how specialized occupational knowledge can support establishment-level productivity, even when establishments are not at the frontier of formal innovation.

3 Subsector-Occupation Complementarity

The primary goal of this paper is to estimate the average effect of an establishment's subsector-occupation complementarity—measured through an establishment-specific score—on establishment size, controlling for lagged size. Various specifications will be explored, involving alternative choices of controls, fixed effects, and interaction terms.

Subsector-occupation complementarity is measured using a revealed comparative advantage (RCA) index adapted to occupations and subsectors. Traditionally, RCA has been used to capture a country's comparative advantage in producing specific goods for export. Here, it is employed to quantify the strength of alignment between specific occupations and subsectors within the labor market. Before providing the formal construction of the complementarity score, it is useful to offer an intuitive overview of the concept I aim to target.

Ideally, one would like to establish a causal relationship between an occupation's presence in an establishment and the establishment's productivity within a given subsector. That is, certain occupations may be inherently more complementary to the production processes characteristic of specific subsectors. Establishments employing such occupations more intensively may be more productive than otherwise similar establishments.

As noted earlier, the subsector-occupation complementarity score used here builds on the RCA index of Balassa (1965) and its application to labor markets in Jara-Figueroa et al. (2018). While Jara-Figueroa et al. (2018) use this measure primarily as a robustness check for assessing firm growth via worker knowledge flows, this paper repurposes it as a direct measure of establishment-level occupational specialization. The focus here is on the most granular industrial classification—subsectors—though parallel measures using broader sectoral and industry classifications are also constructed for robustness.

[1] By "innovation," I refer specifically to formal innovations such as patents or new product inventions, as opposed to process improvements.

3.1 Constructing the Subsector-Occupation Complementarity Score

The construction of the complementarity score proceeds as follows. Define $\phi^t_{s,o}$ as:

$$\phi^t_{s,o} = \left(\frac{L^t_{s,o}}{L^t_o}\right) \Big/ \left(\frac{L^t_s}{L^t}\right) \tag{2}$$

where:

- $L^t_{s,o}$ is the number of workers in year t employed in subsector s and occupation o.
- L^t_o is the total number of workers in occupation o in year t.
- L^t_s is the total number of workers in subsector s in year t.
- L^t is the total number of workers in the economy in year t.

Occupations are classified using the CBO 1994 system. L^t is calculated from the RAIS data after appropriate cleaning (as discussed in Section 4). Establishments and employees are matched by municipality and subsector. Not all establishments are matched individually; rather, they are grouped into establishment groups defined by a subsector-municipality pair. These pairs form the basis of the analysis.

Three different sectoral classifications are available: subsectors (most granular), two-digit subsectors (intermediate), and industries (broadest). Let Sub denote the set of subsectors, $Sub2$ denote the set of two-digit subsectors, and Ind denote the set of industries. For any establishment group g, let $s(g)$, $s2(g)$, and $i(g)$ denote the associated subsector, two-digit subsector, and industry, respectively. Let M denote the set of municipalities in the full data sample, O the set of occupations in the full sample, and $G = Sub \times M$ the set of establishment groups in the full sample.

The establishment-level complementarity scores are then defined as weighted averages of ϕ-scores across occupations:

$$Sub_Occ^t_g = \sum_{o \in O} \left(\frac{L^t_{o,g}}{L^t_g}\right) \phi^t_{s(g),o} \tag{3}$$

$$Sub2_Occ^t_g = \sum_{o \in O} \left(\frac{L^t_{o,g}}{L^t_g}\right) \phi^t_{s2(g),o} \tag{4}$$

$$Ind_Occ^t_g = \sum_{o \in O} \left(\frac{L^t_{o,g}}{L^t_g}\right) \phi^t_{i(g),o} \tag{5}$$

where:

- $L^t_{o,g}$ is the number of workers in occupation o within group g at time t.
- L^t_g is the total number of workers in group g at time t.

Thus, each complementarity score reflects the weighted average of occupation-specific RCA values, with weights given by the employment shares of each occupation within the establishment group.

Intuitively, a high subsector-occupation complementarity score indicates that an establishment group hires occupations that are disproportionately concentrated in its subsector relative to the economy as a whole. If an occupation's $\phi^t_{s,o}$ score equals 1 across all subsectors, that indicates that occupation o's employment distribution matches the economy-wide subsector distribution. A score greater than 1 indicates relative specialization in a particular subsector for occupation o.

The concept of subsector-specific specialization is central: an establishment is "specialized" if its occupational mix aligns with the comparative advantage patterns of its subsector. Nonetheless, this measure has important limitations. Consider the example of lawyers and doctors. If lawyers are employed across a wider range of sectors than doctors, lawyers may appear less specialized relative to their core industry (law), even if they are indispensable within law firms. The ϕ-score abstracts from the extent to which occupations are demanded across subsectors for reasons unrelated to their intrinsic complementarity. This point will be given greater clarity in the next section.

Thus, the following question is raised: to what extent does observed occupational concentration genuinely reflect underlying complementarity rather than broader labor market dynamics? Subsection 3.2 develops a theoretical framework, drawing on Eaton & Kortum (2002) and French (2017), to provide a more rigorous definition of the terms "comparative advantage" and "complementarity" within the framework employed in this paper. Once the definitions are set, the conditions are formalized under which subsector-occupation complementarity can be interpreted as a meaningful indicator of comparative advantage.

3.2 Modeling Revealed Comparative Advantage

3.2.1 French's Model

The subsector-occupation complementarity score used in this paper is an adaptation of the revealed comparative advantage index originally proposed by Balassa (1965). While Balassa's application was in international trade—comparing countries and products—this paper applies the framework to occupations and subsectors of the economy. It is thus necessary to consider under what conditions the Balassa index accurately captures revealed comparative advantage in this new setting. To explore these conditions, I build on the extension of the Eaton & Kortum (2002) model developed by French (2017). French adapts the Eaton-Kortum framework to a multi-sector setting where each country produces a continuum of products, and each product is traded internationally. Within French's framework, the Balassa index provides an accurate measure of comparative advantage when trade and transportation frictions are relatively low.

The model developed here further extends French's approach, adapting it to labor markets where occupations, rather than products, form the focus of the analysis, and municipalities, rather than countries, are the geographic unit. Establishments within each subsector still seek to minimize production costs to meet a fixed level of product demand, just like in French's model. However, labor inputs are now occupational roles, and employing labor may involve hiring from different subsectors and, when necessary, paying to retrain them. The framework embeds these labor market dynamics into the existing structure proposed by French (2017).

Before introducing the extensions, it is useful to briefly summarize key features of French's model. In his setup, the marginal cost for a country i to produce a good of variety ω in subsector j and product category k, and to ship it to country n, is given by:

$$c_{n,i}^{j,k}(\omega) = \frac{c_i^j d_{n,i}^{j,k}}{z_i^{j,k}(\omega)} \tag{6}$$

where:

- c_i^j is the unit input cost in sector j in country i.
- $d_{n,i}^{j,k} \geq 1$ is the iceberg trade cost between countries i and n.
- $z_i^{j,k}(\omega)$ is a random variable capturing the productivity with which inputs are transformed into outputs for product (j, k, ω).

The productivity term $z_i^{j,k}(\omega)$ follows a Fréchet distribution:

$$F_i^{j,k}(z) = e^{-T_i^{j,k} z^{-\theta^j}} \tag{7}$$

where $T_i^{j,k}$ is a technology parameter that determines the overall level of productivity for producing goods of type (j, k) in country i, and $\theta^j > 1$ governs the dispersion of productivity.

In the extension developed here, the basic production structure remains intact, but the interpretation shifts. The index i now refers to municipalities rather than countries, and subsector j is assumed to produce a single good. The market demand for each good, denoted $\overline{Q_s}$, is exogenously given. Establishments minimize costs to produce $\overline{Q_s}$, and unlike in French (2017), the production input bundle c_i is no longer treated as fixed; instead, it is decided by establishments' hiring decisions over occupational groups.

With these adaptations in place, the model provides a framework for assessing the conditions under which the observed subsector-occupation complementarity score accurately reflects underlying comparative advantage.

3.2.2 Model

In this model, labor supply is assumed to be perfectly elastic across all occupations. Establishments operate within a given subsector and choose the set of occupations to employ. Workers can be hired either from within the same subsector or from different subsectors and occupations. When a worker switches from one occupation or subsector to another, establishments incur a "retraining" cost, modeled as a multiple of the worker's previous wage.

Specifically, if an employee initially belongs to occupation o' and subsector s' and is hired into occupation o and subsector s, the establishment must pay:

$$w_{s'}^{o'} d_{s,s'}^{o,o'}$$

where $w_{s'}^{o'}$ is the worker's prior wage and $d_{s,s'}^{o,o'} \geq 1$ captures the retraining cost. Notably, $d_{s,s}^{o,o} = 1$ for all s and o; there is no retraining cost if the occupation and subsector remain unchanged. An important modeling choice is that retraining costs are based on the worker's former wage rather than their new wage. This feature reflects the underlying assumptions about skill acquisition, worker behavior, and employer evaluation, which are outlined below.

Workers acquire formal occupational training before entering the labor market. During this

process, each worker chooses a "skill diversification strategy," denoted $\zeta \in \Omega$, representing their orientation toward generalization or specialization. The ζ values are distributed uniformly across workers to reflect that workers are indifferent about the choice they make. They are indifferent because, as will be seen, their choice has no effect on productivity or payoff. No ζ value requires any more effort, on the part of workers, compared to any other, and they are not a reflection of some innate disposition. Nevertheless, once chosen, it becomes a permanent characteristic through force of habit and cannot subsequently be changed. Importantly, ζ represents a broad cognitive approach, not detailed technical knowledge, specific work experience, or an underlying cognitive ability. Workers are identical in innate ability and have no preference for any occupation or subsector.

While the interpretation of ζ described above will be used throughout the exposition of this model, it should be noted that the functional role that ζ plays in this model is that it represents the diversity of worker types within an occupation. One can similarly think of a model where ζ represents a worker's personality or proclivity for a certain work style, and ζ would still play the same role within that model as it does here where it represents a "specialization type". As long as each worker-type is no more productive than any other, in an ex ante sense, the functional role of ζ is the same. The practical implication of having an infinite spectrum of worker types within occupations is the same as the implication of having "varieties" of products in Eaton & Kortum (2002). It formalizes the idea that there can be within-occupation job flows between subsectors of the economy even though certain subsectors have a comparative advantage in hiring a given occupation.[2]

Each subsector embeds workers within a distinct technological milieu and knowledge environment, reflecting recent innovations in process management (rather than product innovation). Establishments in this model do not create new products; they focus on improving the efficiency with which existing products are produced. Employees, upon entry, must devise ideas about how to apply their skills within the local production environment. The hiring process unfolds in two stages:

1. Initially, employees are hired into the occupations for which they were formally trained. In the first period, no poaching across subsectors is possible.

2. Subsequently, once everyone has been hired, establishments can hire workers across occupations and subsectors. In this stage, candidates propose how they would apply their skills to the target position.

After the first stage, workers of every ζ-type are employed across all $O \times S$ occupation-subsector pairs. In the second stage, establishments interview candidates across the full set of $O \times S$ possibilities for each ζ-type, selecting the applicant whose approach they judge to have the highest productivity. Productivity is thus a joint outcome of formal skills and the quality of the worker's proposed application of those skills. Furthermore, this process may entail the evaluation of each applicant's ideas of how to apply skills through tests or a training period of some kind. An important caveat to this process is that it is assumed to be costless for the employer. The employer is allowed a sneak peak into how employees apply their skills (some of which would be taught for the explicit purposes

[2]The parallel in Eaton & Kortum (2002) is that there can still be a scenario where all countries produce and trade a certain product even though only a single country has a comparative advantage in producing the bulk of varieties of that product.

of this evaluation) without having to pay the retraining cost associated with hiring the applicants.[3] Most importantly, this "sneak peak" gives employers full knowledge of the productivity level each applicant will have if employed.

To make sense of this idea, it should be noted that there is a discrepancy in the knowledge that the hiring managers have and the knowledge that the applicants have. The hiring managers can accurately assess which approaches will lead to higher productivity because they have a fuller understanding of the subsector-specific knowledge and technological milieu than the applicants have. This fuller understanding is noncommunicable and takes time to develop, which prevents the hiring managers from communicating it to each applicant. Applicants are being assessed on the tacit knowledge they display, a form of knowledge which is also noncommunicable (but can be observed). Hiring managers are able to observe it and can accurately assess the long-run productivity it will lead to within the subsector-specific knowledge and technological milieu in which the applicants will be working. It is precisely this broader gestalt-level awareness of the work environment which is key to knowing whether, within that environment, a certain application of skills (existing in the form of tacit knowledge) will be more (or less) productive than any other application of skills. However, due to the noncommunicable nature of tacit knowledge, hiring managers can not instruct workers on how to come up with the tacit knowledge appropriate for that work environment. They can merely select the workers who already demonstrate having it. Refer to Subsection A.1 for a more elaborated description of the intricacies of this process.

Labor markets are assumed to be perfectly competitive. There exists a large pool of unemployed individuals willing to undertake occupational training for any occupation, ensuring that no worker possesses market power. Wages reflect reservation values, expenses incurred from occupational training, and the psychic cost of on-the-job training, rather than bargaining outcomes. Note that the psychic cost of on-the-job training does not vary from individual to individual. It only varies from one occupation-subsector environment to another.

More precisely, wages are composed of three components:

$$w_s^o = w + w_o + w_{s,o}$$

where w captures the general compensation for labor effort, common to all workers (i.e., a reservation wage), w_o reflects the cost (effort, time, resources) of acquiring formal occupational training, and $w_{s,o}$ compensates for the additional effort required by occupation o, in subsector s, to acquire the on-the-job skills necessary for job competency.

The term $w_{s,o}$ deserves particular attention. While formal skills are acquired through structured training, tacit knowledge—unwritten, experience-based understanding—can only be acquired through immersion within a specific subsector. Much has been written about knowledge that can not be acquired by formal training alone, known by different names, such as tacit knowledge (Polanyi, 1958) or "learning by doing" (Bessen, 2015). This form of knowledge is occupation and subsector-specific since subsector environments differ in how easily they allow occupations to

[3]There are clearly costs associated with assessing productivity which are not trivial to real employers even though the process is assumed to be costless in this model. Furthermore, there are numerous other frictions associated with searching and matching which get ignored in this model. All of these frictions are ignored because the purpose of this model is to give an illustration of the concept of subsector-occupation complementarity. While frictions inherent in the process of identifying talented workers and hiring them are of great consequence to real job markets, they are tangential to the concept being illustrated here.

acquire this tacit knowledge.

An important distinction is made in this model between skills that can be acquired through equal effort by all worker, in so far as they are in the proper occupation-subsector environments, and productivity from idea generation. The term $w_{s,o}$ represents on-the-learning. It is tacit knowledge because it is not communicable. It can only be acquired through observation and immersion into a specific role and environment. However, everyone is able to acquire this knowledge. When it is harder to attain, workers can only be incentivized to acquire it through higher compensation.

In subsectors where the environment is more conducive to rapid tacit knowledge acquisition (e.g., strong mentoring, established knowledge ecosystems), workers require less effort to reach full productivity and can therefore be paid lower wages. Conversely, in more challenging environments, higher wages are needed to compensate workers for the additional effort required. There is another component to tacit knowledge which comes from idea generation, a component which is entirely due to luck. The term $w_{s,o}$ does not represent that component. It represents the learnable component.

These dynamics create incentives for cross-subsector hiring. Let's consider an example where there could be an incentive for cross-subsector hiring, and, let's exclude the possibility of cross-occupation hiring to simplify the example. Suppose a worker in occupation o gains tacit skills easily in subsector s' (e.g., tech) and is then hired into subsector s (e.g., finance). The finance employer can choose to either: (1) hire a worker for occupation o with no experience in subsector s, paying a higher wage to compensate for difficult tacit knowledge acquisition; or (2) poach a worker from s' who already possesses closely related tacit skills, paying their former wage plus a retraining cost.

Retraining costs, $d^{o,o'}_{s,s'}$, capture the additional effort and adjustment required when transferring tacit knowledge from one setting to another. Crucially, because tacit skills are formed relative to the former environment, the retraining cost is naturally modeled as a multiple of the worker's prior wage, not their future wage. It is important to remember that employees have no market power, and the wage afforded to them is purely a function of the effort and money required for occupational training and tacit skills acquisition. Moreover, employers are completely aware of each worker's market wage based on those factors.

Poaching will be attractive when:

$$d^{o,o}_{s,s'}(w + w_o + w_{s',o}) < w + w_o + w_{s,o}$$

which can be rearranged as:

$$(d^{o,o}_{s,s'} - 1)(w + w_o) < w_{s,o} - d^{o,o}_{s,s'} w_{s',o}$$

In words, poaching is more likely when retraining costs are low and when the difficulty of acquiring tacit knowledge in the destination subsector from scratch ($w_{s,o}$) exceeds the effective effort needed to transfer existing tacit knowledge from the source subsector to the destination subsector ($d^{o,o}_{s,s'} w_{s',o}$).[4] Thus, cross-subsector hiring patterns reflect underlying differences in the ease of tacit knowledge acquisition across environments, rather than differences in formal skill acquisition.

[4]It should be noted that the effort needed for on-the-job training does not get easier over time within the same occupation. The term "training" should be interpreted more broadly. One is always training to improve one's skills, and, in a dynamic setting, improvements are always needed. Therefore, $w_{s',o}$ being lower than $w_{s,o}$ is to say that the style of learning acquired from subsector s' (a style which can not be learned or communicated outside of subsector s') can be brought over to the destination subsector s after it is learned in subsector s'. It does not represent a static set of skills, but

Establishments can also engage in cross-occupational hiring, in which case they have to pay a retraining cost associated with retraining formal skills.

Ultimately, hiring decisions depend not only on wages and retraining costs, but also on the productivity that each worker offers within the target occupation and subsector. Let $z^{o}_{i,s}(\zeta)$ denote the productivity associated with idea-generation of a worker with skill diversification ζ when employed in occupation o in subsector s within municipality i. Note that this form of productivity is a product of chance. It can not be learned through observation and the ideas in question are not learnable through communication. Using this notion of productivity, the cost of hiring a worker into occupation o and subsector s from origin occupation o' and subsector s' is given by:

$$c^{o,o'}_{i,s,s'}(\zeta) = \frac{w^{o'}_{i,s'} d^{o,o'}_{i,s,s'}}{z^{o,o'}_{i,s,s'}(\zeta)} \tag{8}$$

where:

- $w^{o'}_{i,s'}$ is the wage from the worker's origin occupation and subsector.
- $d^{o,o'}_{i,s,s'} \geq 1$ is the retraining cost multiplier.
- $z^{o,o'}_{i,s,s'}(\zeta)$ is the realized productivity associated with transferring the worker for the new role.

Higher productivity lowers the effective cost of employment, as greater output per unit of wage and retraining cost is achieved. Conceptually, hiring costs should be understood in terms of the cost per unit of labor output, not merely the wage expenditure. Importantly, while $w^{o'}_{i,s'}$ and $d^{o,o'}_{i,s,s'}$ are fixed parameters, the productivity term $z^{o,o'}_{i,s,s'}(\zeta)$ is a random variable. It is drawn from a Fréchet distribution, following Eaton & Kortum (2002) and French (2017):

$$F^{o}_{i,s}(z) = e^{-T^{o}_{i,s} z^{-\theta^{s}}} \tag{9}$$

where:

- $T^{o}_{i,s} > 0$ governs the average productivity level. It represent productivity attributable to ideas, namely ideas of how to apply skills.
- $\theta^{s} > 1$ governs the dispersion of productivity across workers within subsector s.

A higher $T^{o}_{i,s}$ implies that occupation o is more complementary to the technological and knowledge environment of subsector s, leading to systematically higher productivity draws. Meanwhile, a higher θ^{s} implies lower variance in productivity among workers within subsector s. For instance, an AI specialist would exhibit a higher $T^{o}_{i,s}$ when matched with an AI software subsector compared to a general computer scientist, reflecting deeper complementarities. As of now, notions of opportunity cost have not been introduced. However, in the more rigorous formalization of "complementarity" that is to come, notions of opportunity cost will also play a role. Following the structure of French (2017), but adapted to the labor market context, these productivity differences underpin patterns of specialization and cross-subsector labor movements.

a dynamic process of learning which, given some mentoring and retraining, can be made appropriate for a new setting.

As in French (2017), product markets are perfectly competitive. This paper extends the framework by modeling the labor market dynamics between occupations and subsectors, but maintains the assumption of perfect competition. Workers are price takers, and establishments cannot set wages below the reservation wage w_s^o. The differentiation in worker productivity arises not from monopolistic competition, but from heterogeneity in ideas, strategies, and skill applications, analogous to differentiated varieties in Eaton & Kortum (2002).

Given these assumptions, for each $\boldsymbol{\zeta}$-type, a subsector s seeking to fill occupation o will select the worker with the lowest realized cost across all origin subsector-occupation pairs. That is:

$$\overline{c_{i,s}^{o}}(\boldsymbol{\zeta}) = \min_{s',o'} \left\{ c_{i,s,s'}^{o,o'}(\boldsymbol{\zeta}) \right\} \tag{10}$$

where $\overline{c_{i,s}^{o}}(\boldsymbol{\zeta})$ denotes the minimum cost achieved across all available candidates for the position. Establishments observe both the skill diversification $\boldsymbol{\zeta}$ and the realized productivity $z_{i,s,s'}^{o,o'}(\boldsymbol{\zeta})$ for each candidate, ensuring that the minimum cost can be determined.

To characterize the distribution of $\overline{c_{i,s}^{o}}(\boldsymbol{\zeta})$, I first derive the probability that the cost of hiring from a specific origin (o', s') is below some threshold $\bar{c}$. Starting from the definition:

$$z_{s,s'}^{o,o'}(\boldsymbol{\zeta}) = w_{s'}^{o'} d_{s,s'}^{o,o'} (c_{s,s'}^{o,o'}(\boldsymbol{\zeta}))^{-1} \tag{11}$$

it follows that:

$$c_{s,s'}^{o,o'}(\boldsymbol{\zeta}) < \bar{c} \quad \text{if and only if} \quad z_{s,s'}^{o,o'}(\boldsymbol{\zeta}) > \frac{w_{s'}^{o'} d_{s,s'}^{o,o'}}{\bar{c}}$$

Thus, the cumulative distribution function for the hiring cost from (o', s'), denoted $G_{s,s'}^{o,o'}(\bar{c}, \boldsymbol{\zeta})$, satisfies:

$$G_{s,s'}^{o,o'}(\bar{c}, \boldsymbol{\zeta}) = \mathbb{P}\left(c_{s,s'}^{o,o'}(\boldsymbol{\zeta}) < \bar{c} \right) = 1 - e^{-T_s^o (w_{s'}^{o'} d_{s,s'}^{o,o'})^{-\theta^s} \bar{c}^{\theta^s}} \tag{12}$$

Notice that $G_{s,s'}^{o,o'}(\bar{c}, \boldsymbol{\zeta})$ does not depend on $\boldsymbol{\zeta}$, confirming that workers' initial choice of skill diversification does not affect the distribution of productivity realizations ex ante. For simplicity, I henceforth drop $\boldsymbol{\zeta}$ from the notation, writing:

$$G_{s,s'}^{o,o'}(\bar{c})$$

Since employees draw their productivity realizations independently across subsector-occupation pairs, the probability that the minimum cost across all s' for a given o' is below $\bar{c}$ is:

$$G_s^{o,o'}(\bar{c}) = 1 - \prod_{s'} \left(1 - G_{s,s'}^{o,o'}(\bar{c}) \right) = 1 - e^{-\phi_s^{o,o'} \bar{c}^{\theta^s}} \tag{13}$$

where:

$$\phi_s^{o,o'} = \sum_{s'} T_s^o (w_{s'}^{o'} d_{s,s'}^{o,o'})^{-\theta^s} \tag{14}$$

Extending further, the probability that the minimum cost across all o' and s' is below $\bar{c}$ is:

$$G_s^o(\bar{c}) = 1 - \prod_{o'} \prod_{s'} \left(1 - G_{s,s'}^{o,o'}(\bar{c})\right) = 1 - e^{-\phi_s^o \bar{c}^{\theta^s}} \tag{15}$$

where:

$$\phi_s^o = \sum_{o'} \sum_{s'} T_s^o (w_{s'}^{o'} d_{s,s'}^{o,o'})^{-\theta^s} = T_s^o \sum_{o'} \sum_{s'} (w_{s'}^{o'} d_{s,s'}^{o,o'})^{-\theta^s} \tag{16}$$

These expressions define the full distribution of minimum hiring costs faced by establishments seeking to fill occupation o within subsector s. With the cumulative distributions characterized, I now define two cost indices:

1. The cost of employing workers of occupation o in subsector s, C_s^o.
2. The aggregate cost index for subsector s, C_s.

First, the occupation-specific cost index:

$$C_s^o = \left(\int_\Omega \left(\overline{c_s^o}(\zeta) \right)^{1-\eta_s^o} d\zeta \right)^{\frac{1}{1-\eta_s^o}} \tag{17}$$

where η_s^o denotes the elasticity of substitution across workers with different specialization strategies (i.e., different worker types) within occupation o in subsector s. Then, the aggregate subsector cost index:

$$C_s = \left(\sum_o (\tilde{\beta}_s^o)^{\sigma_s} (C_s^o)^{1-\sigma_s} \right)^{\frac{1}{1-\sigma_s}} \tag{18}$$

where σ_s is the elasticity of substitution across occupations within subsector s, and $\tilde{\beta}_s^o$ are demand shifters. These cost indices summarize the minimum expected costs of assembling a labor force with heterogeneous specialization profiles to produce a given quantity of output. The integrability of $\overline{c_s^o}(\zeta)$ follows from the Fréchet distributional assumptions, ensuring that the integral defining C_s^o is well-behaved. Specifically, it can be shown (see Subsection A.2) that:

$$C_s^o = \left(\phi_s^o\right)^{-\frac{1}{\theta^s}} \left(\Gamma\left(\frac{\theta^s + 1 - \eta_s^o}{\theta^s} \right) \right)^{\frac{1}{1-\eta_s^o}} \tag{19}$$

where $\Gamma(\cdot)$ is the gamma function. With these indices in place, the establishment's problem within each subsector s is to minimize total labor costs, subject to meeting the fixed level of output demand $\overline{Q_s}$. The cost minimization problem is:

$$\min_{\{l_s^o(\zeta) \,|\, \forall o,\, \zeta \in \Omega\}} c_s = \sum_o \int_\Omega \overline{c_s^o}(\zeta) l_s^o(\zeta)\, d\zeta$$

subject to:

$$\left(\sum_o \tilde{\beta}_s^o (L_s^o)^{\frac{\sigma_s - 1}{\sigma_s}} \right)^{\frac{\sigma_s}{\sigma_s - 1}} \geq \overline{Q_s}$$

where:

$$L_s^o = \left(\int_\Omega \left(l_s^o(\zeta) \right)^{\frac{\eta_s^o - 1}{\eta_s^o}} d\zeta \right)^{\frac{\eta_s^o}{\eta_s^o - 1}} \tag{20}$$

L_s^o denotes the effective labor input provided by occupation o to subsector s. Solving this problem (see Subsection A.3) yields the familiar result that each occupation is a function of the relative cost of employing that occupation, i.e., the occupation-subsector-specific labor demand equation:

$$L_s^o = \overline{Q_s} \left(\frac{C_s^o}{\tilde{\beta}_s^o C_s} \right)^{-\sigma_s} \tag{21}$$

where more costly occupations are relatively less employed, and the elasticity of substitution across occupations governs the sensitivity of this allocation.

Using the framework discussed so far, some definitions are in order. The notion of comparative advantage in this model is going to involve a hypothetical state where establishments must do their hiring in a state of subsector-occupational autarky. The term "subsector-occupational autarky" will denote a setting where establishments can (1) only hire workers to fill occupational roles for which they were initially trained and (2) prohibited from poaching workers from other subsectors.[5] Such a notion can only be made complete through equations. Let the parameter $\widetilde{\phi_s^o}$ represent the parameter ϕ_s^o in equation (16) under subsector-occupational autarky:

$$\widetilde{\phi_s^o} = T_s^o (w_s^o)^{-\theta^s} \tag{22}$$

Essentially, it is (16) with each of the retraining costs equal to infinity (representing autarky) except for the case when $s = s'$ and $o = o'$, in which case $d_{s,s'}^{o,o'} = 1$ as a baseline assumption, which leads to the equation shown in (22). Using $\widetilde{\phi_s^o}$, the cost of employment indices $\widetilde{C_s^o}$ and $\widetilde{C_s}$ will denote the cost of employment under the setting of subsector-occupational autarky, both of which are defined as follows:

$$\widetilde{C_s^o} = \left(\widetilde{\phi_s^o} \right)^{-\frac{1}{\theta^s}} \left(\Gamma \left(\frac{\theta^s + 1 - \eta_s^o}{\theta^s} \right) \right)^{\frac{1}{1-\eta_s^o}} \tag{23}$$

$$\widetilde{C_s} = \left(\sum_o (\tilde{\beta}_s^o)^{\sigma_s} (\widetilde{C_s^o})^{1-\sigma_s} \right)^{\frac{1}{1-\sigma_s}} \tag{24}$$

Using the definitions provided, the concept of comparative advantage, within the framework of this model, can be formalized. Drawing from French (2017) and Haberler (1930), the definition of comparative advantage employed here is grounded in a state of autarky and is based on the concept of opportunity cost (in the context of cost of employment).[6] Initially, I will illustrate the definition of comparative advantage using two occupations, o and o', and two subsectors, s and s'. Following this, I will introduce a more general formulation of comparative advantage that involves a comparison

[5] For a detailed explanation of this process, refer to Subsection A.1.

[6] Bernhofen (2005) provides a useful summary of Haberler (1930).

between two occupations, incorporating the notion of opportunity cost across all subsectors.

I begin with the simpler case of comparative advantage. An establishment in subsector s is said to have a comparative advantage in employing occupation o over occupation o', relative to an establishment in subsector s', if

$$\frac{\widetilde{C_s^o}}{\widetilde{C_s^{o'}}} < \frac{\widetilde{C_{s'}^o}}{\widetilde{C_{s'}^{o'}}}. \tag{25}$$

This definition compares two ratios. On the left-hand side, the ratio represents the cost of employing occupation o in subsector s relative to the cost of employing occupation o' in the same subsector. Interpreting the cost of employing o' as an opportunity cost, the measure evaluates the relative expense of employing o in terms of the forgone alternative—employing o'. When this ratio is lower in subsector s than in subsector s', subsector s has a comparative advantage in employing o, reflecting a more favorable trade-off relative to its opportunity cost.

Building on this foundation, I formalize the concept of "complementarity." An establishment in subsector s exhibits greater complementarity with occupation o relative to occupation o', compared to an establishment in subsector s', if

$$\frac{\left(\frac{T_s^o}{(w_s^o)^{\theta_s}}\right)}{\left(\frac{T_s^{o'}}{(w_s^{o'})^{\theta_s}}\right)} > \frac{\left(\frac{T_{s'}^o}{(w_{s'}^o)^{\theta_{s'}}}\right)}{\left(\frac{T_{s'}^{o'}}{(w_{s'}^{o'})^{\theta_{s'}}}\right)}. \tag{26}$$

Recall that the wage w_s^o represents the reservation wage for workers, augmented by returns to occupational training, and compensation for on-the-job effort. Wages capture productivity unrelated to idea generation, while T_s^o measures productivity arising specifically from idea-generation and innovation. Consequently, the ratio $\frac{T_s^o}{(w_s^o)^{\theta_s}}$ can be interpreted as a productivity measure: it captures the value of idea-generated productivity net of the cost of labor.

The parameter θ_s warrants further discussion. Recall that z follows a Fréchet distribution with mean T_s^o and variance determined by θ_s. A smaller θ_s implies a higher variance, producing a thicker right tail. Since workers from different occupation–subsector pairs compete for each job, and the applicant with the highest realized z is hired, a thicker tail increases the expected productivity of the selected worker. Thus, as variance rises, $\frac{T_s^o}{(w_s^o)^{\theta_s}}$ increases, reflecting the higher expected realized productivity generated by the interaction between occupational skills and subsector-specific environments.

Complementarity should therefore be understood as the enhanced productivity, beyond wage costs, arising from the fit between a particular occupation and the subsector environment. This measure is always evaluated relative to an alternative scenario (here, occupation o') and compared across subsectors. Higher complementarity thus identifies a comparative advantage in productivity terms, complementing the cost-based notion introduced previously.

With these definitions in place, consider a few simplifying assumptions. Two sufficient conditions allow the following inferences: (1) $\tilde{\beta}_s^o = 1 \quad \forall o, s$, eliminating demand shifters, and (2) $\sigma_s = \sigma_{s'}$, equalizing elasticities of substitution across subsectors. While these assumptions are restrictive, they serve the purpose of isolating complementarity: by holding labor demand parameters constant, observed hiring patterns reflect underlying productivity differences rather than exogenous demand

effects. The second condition, regarding elasticity, is only necessary for this simplified scenario and is relaxed in the broader formulation. Under these conditions, the following proposition holds:

Proposition 1: If (1) and (2) are satisfied, then

$$\frac{L_s^o}{L_s^{o'}} > \frac{L_{s'}^o}{L_{s'}^{o'}} \implies \frac{C_s^o}{C_s^{o'}} < \frac{C_{s'}^o}{C_{s'}^{o'}},$$

and if $\eta_s^o = \eta_s \ \ \forall o$,

$$\frac{\widetilde{C_s^o}}{\widetilde{C_s^{o'}}} < \frac{\widetilde{C_{s'}^o}}{\widetilde{C_{s'}^{o'}}} \implies \frac{\left(\frac{T_s^o}{(w_s^o)^{\theta_s})}\right)^{-\frac{1}{\theta_s}}}{\left(\frac{T_s^{o'}}{(w_s^{o'})^{\theta_s})}\right)^{-\frac{1}{\theta_s}}} > \frac{\left(\frac{T_{s'}^o}{(w_{s'}^o)^{\theta_{s'}})}\right)^{-\frac{1}{\theta_{s'}}}}{\left(\frac{T_{s'}^{o'}}{(w_{s'}^{o'})^{\theta_{s'}})}\right)^{-\frac{1}{\theta_{s'}}}}.$$

The condition $\eta_s^o = \eta_s$ ensures that the capacity for integrating different worker specialization types are uniform across occupations in the same subector. While unlikely to hold universally, it may reasonably apply to a core set of occupations in each subsector. Proposition 1 therefore establishes that, under subsector–occupation autarky, revealed comparative advantage in employment patterns implies complementarity in productivity:

$$\frac{L_s^o}{L_s^{o'}} > \frac{L_{s'}^o}{L_{s'}^{o'}} \implies \frac{\left(\frac{T_s^o}{(w_s^o)^{\theta_s})}\right)^{-\frac{1}{\theta_s}}}{\left(\frac{T_s^{o'}}{(w_s^{o'})^{\theta_s})}\right)^{-\frac{1}{\theta_s}}} > \frac{\left(\frac{T_{s'}^o}{(w_{s'}^o)^{\theta_{s'}})}\right)^{-\frac{1}{\theta_{s'}}}}{\left(\frac{T_{s'}^{o'}}{(w_{s'}^{o'})^{\theta_{s'}})}\right)^{-\frac{1}{\theta_{s'}}}}. \tag{27}$$

See Subsection A.4 for a formal proof. The key caveat is that the link between revealed comparative advantage and productivity-based complementarity only holds under subsector-occupation autarky. This formulation serves to build intuition about the concept, paving the way for a more general treatment.

To extend the notion of opportunity cost beyond a single alternative, consider the problem from the occupation's perspective:

$$\frac{\widetilde{C_s^o}}{\widetilde{C_s^{o'}}} < \frac{\widetilde{C_{s'}^o}}{\widetilde{C_{s'}^{o'}}} \implies \frac{\widetilde{C_s^o}}{\widetilde{C_{s'}^o}} < \frac{\widetilde{C_s^{o'}}}{\widetilde{C_{s'}^{o'}}}. \tag{28}$$

Here, the right-hand side expresses that occupation o has a comparative advantage in being employed in subsector s relative to s', with the forgone opportunity of being employed in subsector s'. More broadly, occupation o may be employed in multiple alternative subsectors, each representing a different opportunity cost. To formalize this broader notion, I construct an index capturing comparative advantage across all feasible subsector allocations.

$$ACD^o = \left(\sum_{s'} \left(\frac{C_{s'}^o}{C_{s'}}\right)^{-\sigma_{s'}} \overline{Q_{s'}}\right)^{-1} \tag{29}$$

The ACD^o measure serves as an index capturing the absolute cost disadvantage (or advantage)

of occupation o across all subsectors of the economy. Here, $C_{s'}$ represents the marginal cost of producing an additional unit of output in subsector s', and $C^o_{s'}/C_{s'}$ denotes the relative cost of employing occupation o compared to the overall cost of production in that subsector. A higher ratio indicates that occupation o is relatively more expensive to employ within that subsector.

The exponent $-\sigma_{s'}$ accounts for the elasticity of substitution between occupations, which varies across subsectors. Intuitively, if $C^o_{s'}/C_{s'}$ is relatively high in one subsector, its effect on ACD^o is amplified when $\sigma_{s'}$ is low. Low $\sigma_{s'}$ implies that occupations are less substitutable, so the establishment must employ occupation o more intensively than in subsectors with higher $\sigma_{s'}$, where substitution is easier. Consequently, ACD^o aggregates cost disadvantages while accounting for differential substitution effects across subsectors.

Each subsector's contribution to the index is weighted by its output, $\overline{Q_{s'}}$, reflecting the greater influence of larger subsectors on aggregate employment patterns. The overall sum is raised to the power -1 to preserve the familiar CES-like structure of price indices, despite the fact that elasticity varies across subsectors. This formulation avoids the need to assign a specific $-\frac{1}{\sigma_{s'}}$ exponent for each subsector individually.

By construction, a higher ACD^o indicates a greater absolute cost disadvantage for occupation o across the economy. The index provides a foundation for identifying comparative advantages of occupations across subsectors, which will be formalized in subsequent propositions.

To complement the cost-based index, I construct a parallel index capturing productivity disadvantages. Define productivity in terms of innovation-adjusted units of wages as $\rho^o_s = T^o_s/(w^o_s)^{\theta_s}$. A productivity disadvantage is then expressed as $(\rho^o_s)^{-\frac{1}{\theta_s}}$, consistent with the Frechet framework where productivity dispersion is determined by the variance parameter θ_s. Analogous to ACD^o, the subsector-level productivity disadvantage index is:

$$APD_s = \left(\sum_{o''} (\rho^{o''}_s)^{-\frac{1}{\theta_{s''}}(1-\sigma_{s''})} \right)^{\frac{1}{1-\sigma_{s''}}} \tag{30}$$

This index captures the absolute productivity disadvantage of subsector s, allowing for the computation of relative disadvantages for each occupation as $(\rho^o_s)^{-\frac{1}{\theta_s}}/APD_s$. Aggregating across subsectors yields:

$$APD^o = \left(\sum_{s'} \left(\frac{(\rho^o_s)^{-\frac{1}{\theta_s}}}{APD_s} \right)^{-\sigma_s} \overline{Q_s} \right)^{-1} \tag{31}$$

The APD^o index thus represents the absolute productivity disadvantage of occupation o across the economy. Using these indices, I formalize the following proposition:

Proposition 2. If $\tilde{\beta}^o_s = 1$ for all o and s, then:

$$\frac{L^o_s/L_s}{L^o/L} > \frac{L^{o'}_s/L_s}{L^{o'}/L} \implies \frac{C^o_s}{(ACD^o)^{1/\sigma_s}} < \frac{C^{o'}_s}{(ACD^{o'})^{1/\sigma_s}}$$

Furthermore, if $\eta^o_s = \eta_s$, then under subsector-occupation autarky:

$$\frac{\widetilde{C_s^o}}{(\widetilde{ACD}^o)^{1/\sigma_s}} < \frac{\widetilde{C_s^{o'}}}{(\widetilde{ACD}^{o'})^{1/\sigma_s}} \implies \frac{(\rho_s^o)^{1/\theta_s}}{(APD^o)^{-1/\sigma_s}} > \frac{(\rho_s^{o'})^{1/\theta_s}}{(APD^{o'})^{-1/\sigma_s}}$$

Subsection A.5 provides the proof.

Proposition 2 restates Proposition 1 in terms of the newly defined indices. The cost indices are normalized by the absolute cost disadvantage, accounting for substitution. When substitutability is high in a subsector, the absolute disadvantage is less relevant, as only occupations with lower employment costs will be chosen, allowing direct comparison of o and o'. Analogously, higher productivity in a subsector relative to the occupation's aggregate productivity indicates a comparative advantage rather than merely an absolute advantage.

It is important to note that this framework assumes representative establishments and workers within each occupation-subsector pair, abstracting from inter-establishment productivity differences. While this simplification does not capture all real-world heterogeneity, it provides intuition for the ideal allocation of occupations to subsectors.

For empirical section to follow, the relevant variation arises from municipality-by-subsector interactions over time. Proposition 2 compares occupations within a subsector, aligning with the regression design, which exploits within-municipality-by-subsector variation. Real-world deviations, such as spatial frictions affecting labor mobility, could introduce differences across establishments; modeling these frictions is a potential extension of this model.

It should be noted that proposition 2 uses a specific occupation-subsector pair. To get the establishment score, a weighted average is taken across occupations. A question some may have is why it should matter that establishments hire workers with comparative advantage. After all, from the standpoint of the establishment doing the hiring, it does not matter whether the worker's opportunity cost, as expressed above, is high or low. While the existing model can not provide an answer to that question, a future extension of the model can break firms up into different capacities for specialization. The idea here is that some firms have the managerial expertise necessary to prune their workforce so that it encompasses a core set of occupations where "core" is defined in terms or relative productivity advantages captured in the manner shown in proposition 2. The caveat is that these firms would make up a small enough fraction of firms to not impact the aggregate measures that define the revealed comparative advantage score.

Furthermore, the more important question, within the context of this paper, regarding the hypothetical firm just described is the extent to which they benefit from density and industrial diversity in regions and the extent to which they agglomerate. This paper does not provide a model framework for considering this firm type, but, the firm type described should be kept in mind in the empirical section that follows. It is in reference to such a firm type that the regressions are meaningful, since they all assume that establishments benefit from hiring workers with a comparative advantage even if those workers belong to occupations that offer no absolute productivity advantages across all subsectors.

4 Data

4.1 Data Sources and Construction

The primary dataset employed in this study is the *Relação Anual de Informações Sociais* (RAIS), a Brazilian formal labor market census administered by the Ministry of Labor and Employment (MTE). All graphs, tables, and regressions are constructed using variables sourced from RAIS. Supplementary datasets are incorporated for three specific purposes: (i) to adjust wages for inflation, (ii) to correct for shifting geographic boundaries over time, and (iii) to distinguish tradable from nontradable subsectors.

To adjust wages, real minimum wage values from Berg (2010) are used, allowing for wages across years to be expressed in multiples of the 1985 minimum wage, thus ensuring temporal comparability. Geographic boundary corrections rely on the *Areas Mínimas Comparáveis* (AMCs) dataset developed by Ehrl (2017), which provides harmonized municipal boundaries over time. Finally, tradable and nontradable industry classifications are adopted from the supplementary material of Dix-Carneiro & Kovak (2017).

4.2 Description of RAIS

RAIS is a near-comprehensive census of Brazil's formal labor market, capturing nearly all registered establishments and employees. Reporting incentives are strong: workers earning less than twice the minimum wage must submit employment records to qualify for social benefits, and employers face fines for misreporting (Dix-Carneiro & Kovak, 2017). Although RAIS excludes informal sector workers—who account for roughly 35% of employees and 40% of GDP (Ulyssea, 2018)—its coverage of the formal sector is extensive.

Two versions of RAIS exist: an identified version and a publicly available de-identified version. This study uses the public version, consisting of establishment-level data aggregated over groups of employees. Geographic identifiers are available at the municipality, microregion, and state levels, with the municipality being the most granular. To ensure consistent geographic coverage over time, observations are collapsed to the AMC level. Each observation corresponds to a subsector-AMC pair.

4.3 Data Cleaning and Matching Procedure

Establishments and employees are matched at the subsector-municipality level. Employee-level variables—particularly subsector and municipality identifiers—enable the grouping of employees to corresponding establishments. This match is performed separately for each year.

Although the data is thorough in scope, being based on near-universal reporting in the formal sector economy, there are a few limitations of using the public version of the data which need to be discussed. Since there is no way to track employee or establishment movements across time using the public data, it is not possible to tell what fraction of establishments (or employees) within each municipality are the same as those of other years. Only aggregate trends across space and time can be observed.

Furthermore, data does not exist for every municipality at every point in time. Although the AMCs are chosen such that boundaries are harmonized across time, it is possible that for some years,

municipalities within them are missing. This is not necessarily a problem when I take average level indicators that do not have a spatial component (in so far as the missing municipalities do not differ too much, on average, from the non-missing ones). When taking spatial variables, like population density, it makes a meaningful difference if all of the municipalities are present or not. As a result, for regressions involving density, values are only calculated if all municipalities are present in the AMC.

Key variables include establishment-level average wages and educational attainment distributions. Wages are expressed as multiples of the contemporaneous minimum wage. Education is reported as categorical dummies across ten brackets; the fraction of employees in each bracket is computed for each subsector-AMC pair. In all regressions, analytical weights proportional to the number of establishments in each subsector-AMC are employed.

5 Main Hypothesis and Empirical Strategy

5.1 Empirical Framework

The core regression specification is:

$$\ln(S_g^t) = \beta_0 + \beta_1 \ln(S_g^{t-1}) + \beta_2 \ln(w_g^t) + \beta_3 \ln(z_g^t) + \sum_j \alpha_j edu_{j,g}^t + \mu_{m,s} + \lambda_t + \epsilon_g^t, \tag{32}$$

where g indexes subsector-AMC pairs and t indexes years.

The dependent variable, S_g^t, represents the average number of employees per establishment. The primary variable of interest, z_g^t, is the subsector-occupation complementarity score. Two variants of the score are analyzed: the traditional RCA measure and the ARCA adjustment proposed by Hoen & Oosterhaven (2006), which offers improved comparability over time.

Control variables include average establishment wages (w_g^t) and the share of employees in each education bracket ($edu_{j,g}^t$). All regressions include AMC-subsector fixed effects ($\mu_{m,s}$) and year fixed effects (λ_t). Analytical weights are the number of establishments in each subsector-AMC.

The coefficient β_3 captures the elasticity of establishment size with respect to the complementarity score, controlling for average wages and educational composition. Because the education controls capture variation in general formal education, but not occupation-specific human capital, β_3 reflects differences in both tacit and specialized formal knowledge associated with occupational backgrounds.

Importantly, controlling for average wages serves to proxy for underlying productivity differences unrelated to specialization—following the rationale of Combes et al. (2011), who argue for wages as a proxy for worker quality in urban productivity studies. Although the primary outcome is establishment size, two additional specifications are estimated to capture other margins of adjustment:

$$\ln(w_g^t) = \beta_0 + \beta_1 \ln(w_g^{t-1}) + \beta_2 \ln(S_g^t) + \beta_3 \ln(z_g^t) + \sum_j \alpha_j edu_{j,g}^t + \mu_{m,s} + \lambda_t + \epsilon_g^t, \tag{33}$$

$$\ln(TW_g^t) = \beta_0 + \beta_1 \ln(TW_g^{t-1}) + \beta_3 \ln(z_g^t) + \sum_j \alpha_j edu_{j,g}^t + \mu_{m,s} + \lambda_t + \epsilon_g^t, \tag{34}$$

where TW_g^t denotes total wages paid by establishments.

These specifications allow for distinguishing whether complementarity leads to larger establishments, higher wages, or higher total wage bills. Differences between tradable and nontradable subsectors are explored to understand how sector characteristics influence these outcomes.

5.2 Summary Statistics and Stylized Facts

Table 1.1 reports mean values of key variables by year, spanning 1985–1993. The average values of the important variables are shown by year in Table 1.1 shown below. Moderate year-to-year variation is evident, with larger fluctuations during periods of macroeconomic instability, particularly 1989–1991 when Brazil was dealing with major inflation.

Table 1.1: Average Values by Year

Variable	**1985**	**1986**	**1987**	**1988**	**1989**	**1990**	**1991**	**1992**	**1993**
Avg Wage	2.391	2.702	2.338	2.411	2.449	1.915	1.789	1.853	2.086
Avg Estab Size (in Employees)	18.646	18.069	18.326	19.066	18.197	16.285	15.747	14.935	14.483
Ave RCA (Subsector)	20.855	19.942	20.238	19.617	18.085	17.585	18.113	18.738	18.386
Avg ARCA (Subsector)	0.058	0.057	0.057	0.058	0.057	0.057	0.060	0.061	0.063

Table 1.2 presents counts of observations, establishments, and subsectors per year. All 228 subsectors are present for every year in the sample. Not all observations are present because the number of AMCs in the sample vary from year to year. Some of this has to do with the fact that the boundaries changed in a way that did not allow all municipalities to belong to any AMC for some years. Nevertheless, there are a substantive number of observations per year.

Table 1.2: Variable Count by Year

Variable	**1985**	**1986**	**1987**	**1988**	**1989**	**1990**	**1991**	**1992**	**1993**
Subsector Count	228	228	228	228	228	228	228	228	228
AMC Count	3 679	3 707	3 788	3 785	3 827	3 676	3 788	3 790	3 829
Observations	102 874	108 251	108 645	108 451	112 425	112 322	114 969	118 805	120 855
Establishment Count	822 131	923 492	926 086	913 818	996 209	1 016 776	1 012 546	1 019 555	1 061 937

Table 1.3 provides summary statistics for core variables. As expected, the standard deviation of the RCA measure is considerably larger than that of the ARCA measure, consistent with the latter's design for symmetry and comparability, as Hoen & Oosterhaven (2006) made clear. With that being said, since the ARCA value that is used here is the establishment ARCA index average rather than the scores given to individual establishments, the average ARCA data points do not form a perfectly symmetric distribution. After applying the natural log to both RCA and ARCA, their distributions become closer to normal. As Table 1.3 shows, the min of average ARCA is -0.059. To deal with the negative terms, I added 0.06 to every point before applying the log transformation.

To demonstrate the reliability of the ARCA measure, I graph the average annual RCA index values and compare it to the average annual ARCA index values. Figure 1 compares annual average RCA values with annual average RCA values using 1985 as the base year. When 1985 is used as the base year, the RCA index takes the ratio of $(L_s^o \ / \ L_s)$ for years 1985-1993 and $(L^o \ / \ L)$ for years 1985. As Figure 1 shows, using 1985 as a base year has some effect on 1990-1994, but 1985-1990 shows a stark reduction regardless of whether 1985 is used as a base year or not.

Table 1.3: Summary Statistics							
Variable	**Mean**	**p50**	**SD**	**Min**	**Max**	**p25**	**p75**
Avg Wage	2.201	1.688	1.731	0	84.171	1.136	2.664
Avg Estab Size (in Employees)	16.986	7.167	58.794	1	13249	4.459	14.772
Avg RCA (Subsector)	18.993	8.553	90.802	0	46591.926	4.004	18.774
Avg ARCA (Subsector)	0.059	0.04	0.06	-0.059	0.651	0.019	0.073
n	783.733	62	1843.163	1	11574	10	447

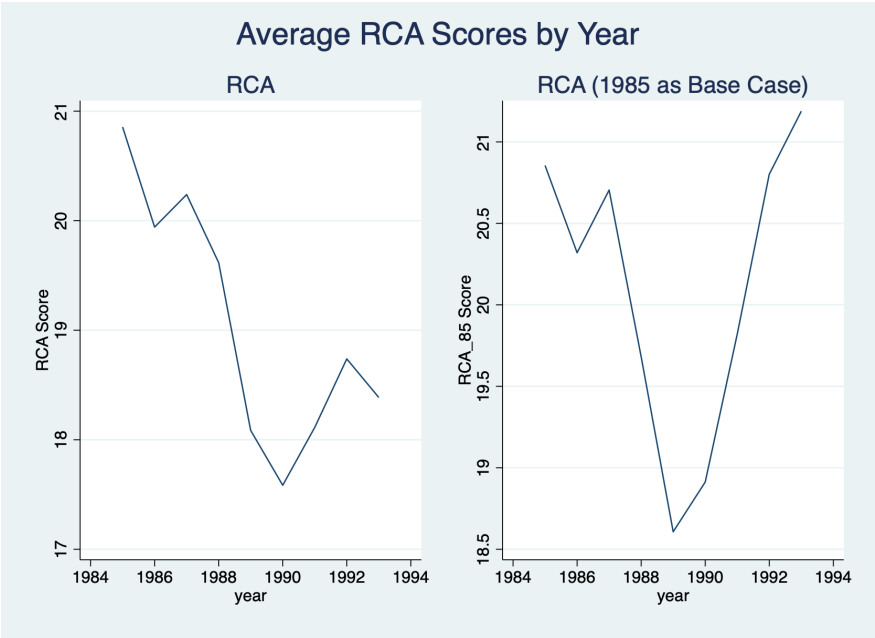

Figure 1: Average RCA Index Values by Year

Figure 2 shows annual average ARCA values. In this case, using 1985 as a base year makes almost no difference. Furthermore, one sees a rise in the ARCA, which is what would be expected if there is a moderate rise in specialization across time. In this case, the difference is not exceptionally large. It goes from .058 to .064. Nevertheless, differences in labor force specialization across time are not going to be very large for a sample of years that is limited in range (1985 to 1993).

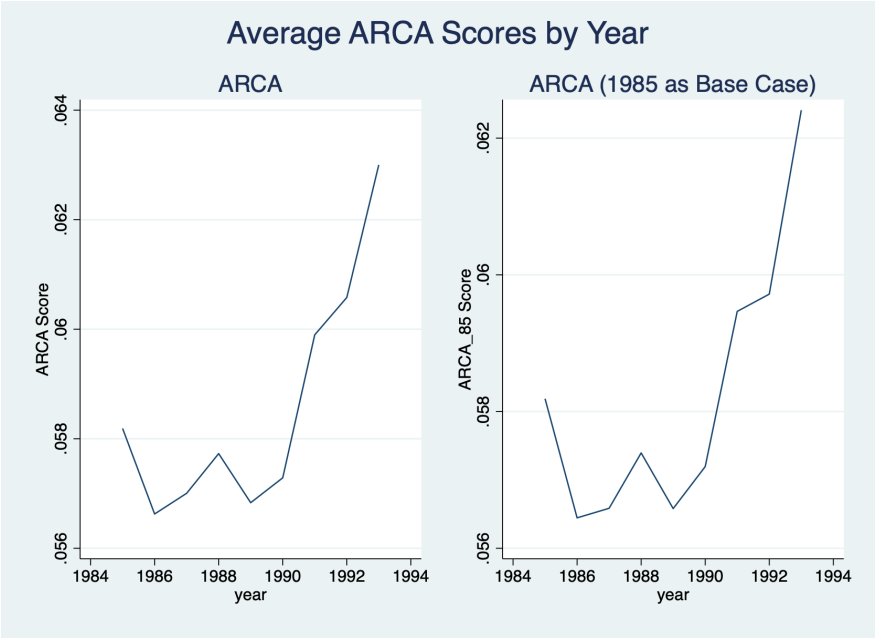

Figure 2: Average ARCA Index Values by Year

Log transformations substantially normalize the skewed distributions of key variables, as Figures 3–6 demonstrate for ARCA, establishment size, density, and wages.

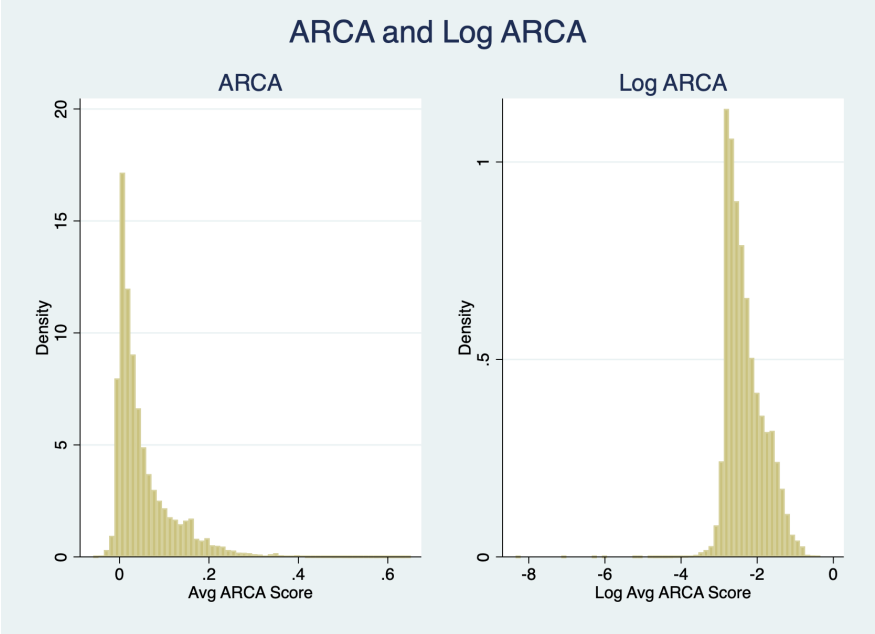

Figure 3: Distribution of ARCA Index (Log vs. Levels)

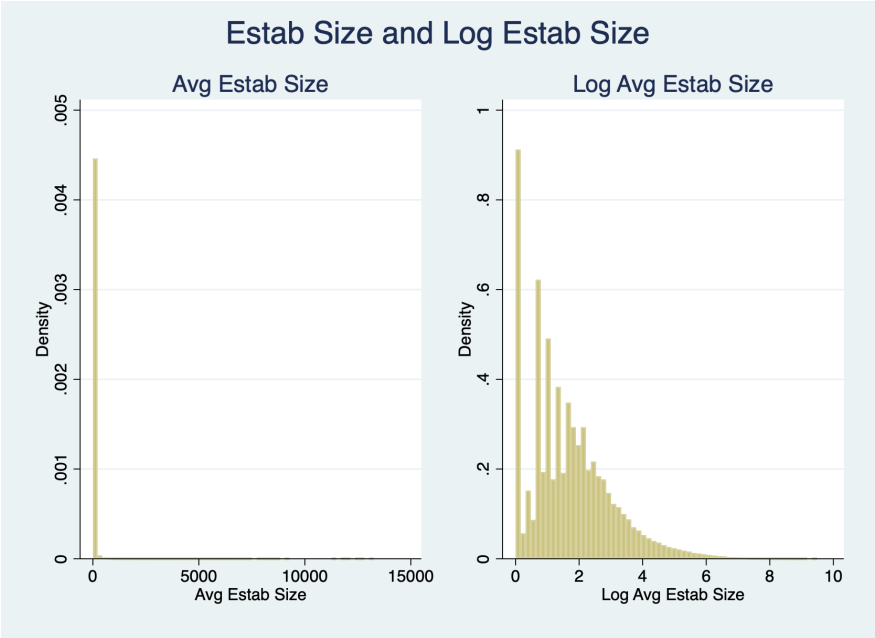

Figure 4: Distribution of Employees per Establishment (Log vs. Levels)

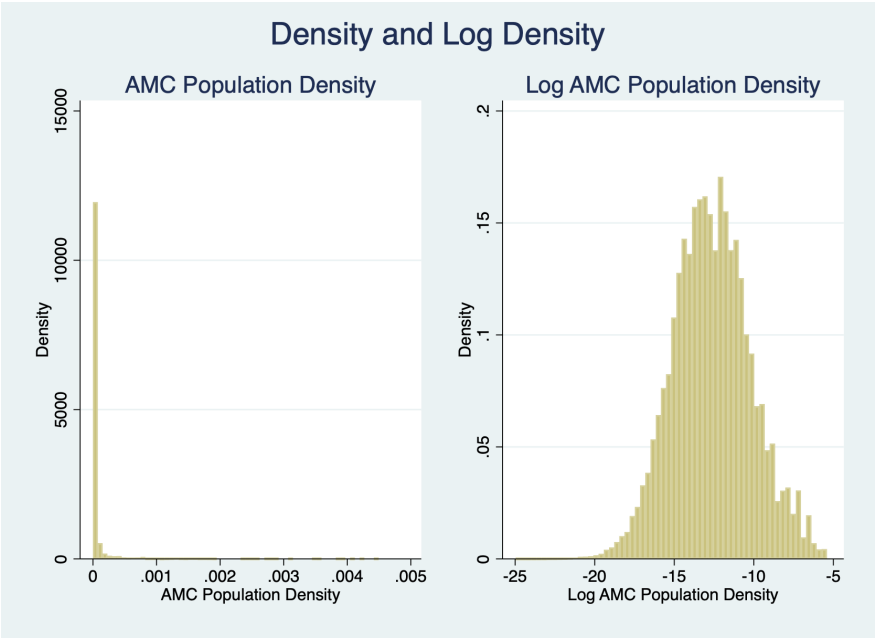

Figure 5: Distribution of AMC Population Density (Log vs. Levels)

Figure 6: Distribution of Average Establishment Wage (Log vs. Levels)

Spatial patterns in complementarity and density are shown in Figures 7 and 8. ARCA levels and density levels vary meaningfully across AMCs, motivating further exploration of density as a moderating factor in the complementarity-growth relationship. The map on the left of Figure 7 shows average subsector ARCA levels, whereas the map on the right of Figure 7 shows each AMC colored by whether they are above or below the median for all AMCs. Purple represents a value below the median whereas yellow represents a value above the median.

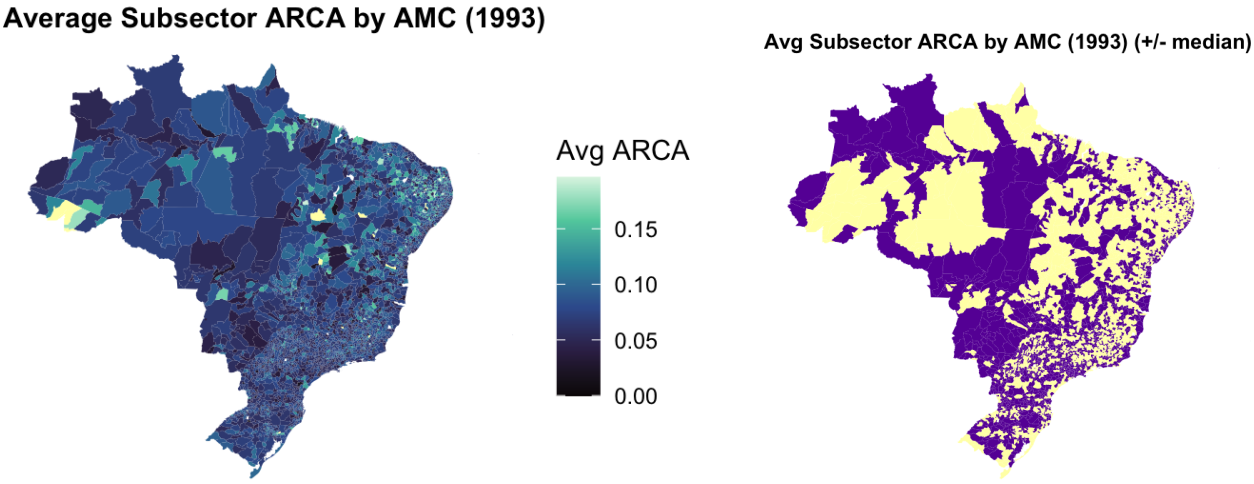

Figure 7: Spatial Distribution of ARCA Index by AMC (1993)

Similarly, a map of average density levels is shown. The density levels are people per square mile. Some of the AMCs are left out because they are part of AMCs that have missing municipalities, and, therefore, the density measures are unreliable. The left hand side has average density levels

mapped. Once again, the right hand side has above-median values in yellow and below-median values in purple.

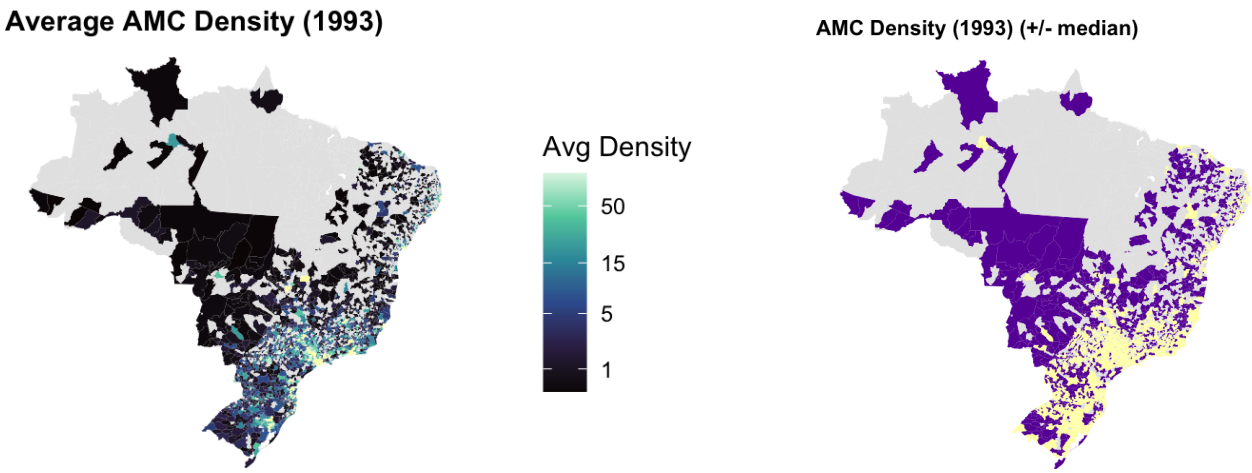

Figure 8: Spatial Distribution of AMC Density (1993). Note: The gray AMCs are those that have missing municipalities. For these, population density is not calculated.

5.3 Motivation

This section motivates the empirical analysis that is to follow, namely researching the relationship between subsector-occupation complementarity and establishment growth, and how this relationship may vary with population density. Before turning to that empirical analysis, it is useful to establish a few basic empirical patterns in the data to properly background the findings to come. The first is the well-documented positive relationship between density and productivity, which holds across all three productivity measures shown in Table 2.1 (which are the three productivity meaures I will use for the main regressions of this paper).

This relationship is demonstrated using a simple OLS regression, where the unit of analysis is the AMC geographic unit—that is, all variables are aggregated at the AMC level. The regressions are weighted by the number of establishments in each AMC. These results are not meant to imply a causal relationship, but rather to document a stylized fact consistent with existing literature. Table 2.1 presents regressions of (i) log average wage, (ii) log average establishment size, and (iii) log total wages on log population density, all of which indicate a positive relationship between density and productivity.

Specifically, a 1% increase in AMC-level population density is associated with a 0.13% increase in average wages per establishment, a 0.212% increase in average establishment size, and a 0.35% increase in total wages per establishment. While subsector composition likely plays a substantial role in driving these patterns, this initial regression includes no controls or fixed effects. At this stage, the goal is not to isolate causal effects, but simply to establish baseline correlations that motivate the analysis to follow.

A second empirical pattern that warrants attention is the negative relationship between density and subsector-occupation complementarity, as measured by the ARCA index. This relationship holds across multiple alternative measures of density. Although the effect is small in magnitude, it is statistically significant and robust. Table 2.2 presents four separate regressions of log average ARCA

Table 2.1: Exploring Density's Effect on Productivity

	Dependent Variable		
	$ln(Avg\ Wage)$	$ln(Avg\ Size)$	$ln(Tot\ Size)$
Model	(1)	(2)	(3)
$ln(density)$	0.139***	0.212***	0.350***
	(0.002)	(0.002)	(0.004)
Constant	1.978***	4.449***	6.423***
	(0.019)	(0.022)	(0.038)
Observations	2,165	2,165	2,165
R-squared	0.712	0.803	0.797
aweights	Yes	Yes	Yes
Level of Observation	AMC	AMC	AMC

Notes: Standard errors in parentheses *** p<0.01, ** p<0.05, * p<0.1

on alternative density measures: (i) log population density at the AMC level, (ii) log establishment density at the AMC level, (iii) log population density of same-subsector employees, and (iv) log population density of same-subsector establishments.

Across all specifications, higher density is associated with slightly lower ARCA scores. For example, a 1% increase in population density is associated with a 0.008% decline in average establishment-level ARCA. Though modest, the estimates are consistently significant at the 1% level. These findings are important because they reveal a negative correlation between density and complementarity, even though both variables are positively associated with productivity, as shown above for density, and as will be shown in the next section for the ARCA index.

The presence of a negative correlation between two variables that are each positively associated with a third is not, by itself, surprising. However, the relationship is notable in this context because it bears directly on a broader question: does urbanization promote or hinder establishment-level specialization? While the available data do not permit a definitive answer, the empirical patterns established here provide meaningful insights and set the stage for the main analysis.

6 Results

6.1 Baseline Findings

I begin by estimating equation (23). The education variables and log wages are included as controls, but, to ensure a more streamlined presentation of the results, estimated coefficient values are not shown for any of the education variables. They are not of primary interest to the analysis done here, unlike the estimated coefficient value of log RCA, which is the main result of interest. Refer to Table 3 for the results.

The results of the regression shown in the first column of Table 3 includes the effect of log

Table 2.2: Exploring Density's Effect on ARCA

	Dependent Variable			
	$ln(Avg\ ARCA)$	$ln(Avg\ ARCA)$	$ln(Avg\ ARCA)$	$ln(Avg\ ARCA)$
Model	(1)	(2)	(3)	(4)
$ln(density)$	-0.008***			
(AMC)	(0.000)			
$ln(estab\ density)$		-0.008***		
(AMC)		(0.000)		
$ln(density)$			-0.007***	
(Subsector-AMC)			(0.001)	
$ln(estab\ density)$				-0.006***
(Subsector-AMC)				(0.001)
Constant	-2.456***	-2.464***	-2.487***	-2.492***
	(0.004)	(0.005)	(0.008)	(0.013)
Observations	2,165	2,165	2,165	2,165
R-squared	0.170	0.137	0.075	0.036
aweights	Yes	Yes	Yes	Yes
Level of Observation	AMC	AMC	AMC	AMC

Notes: Standard errors in parentheses *** p<0.01, ** p<0.05, * p<0.1

average RCA (measured using subsectors) on log average establishment size. Recall the RCA index, unlike the ARCA index, is less appropriate for comparisons across time. Nevertheless, the RCA index is the more well known of the two indices, and it is the index used in the model construction in Subsection 3.2. Therefore, the regressions shown in Table 3, which all use the RCA index, will serve as a good basis of comparison to analyze the results shown in upcoming tables, which use the ARCA index instead of the RCA index.

The results indicate that a one percent rise in average RCA is associated with a 0.014 percent rise in average establishment size. The regression has subsector by AMC fixed effects, and the standard errors are clustered standard errors on the level of AMC by year. Education dummies are included in each of these regressions, although their estimated coefficients are not shown. There are 786,305 observations when one lag term is used in the first and second column, and 632,260 observations when two lag terms are used in column 3. Observations are subsector-AMC pairs. Each subsector-AMC has numerous establishments, and each regression is weighted, using analytical weights, by the number of establishments.

The three regressions shown in Table 3 include log establishment wages, log average RCA, education dummies, and a one-year log establishment size lag term as control variables. In addition to those variables, the regression shown in column two has a one-year log RCA lag term, which

Table 3: Regression of Log Establishment Size on Log RCA

	Dependent Variable		
	$ln(Avg\ Size)$	$ln(Avg\ Size)$	$ln(Avg\ Size)$
Model	(1)	(2)	(3)
$ln(Avg\ Size)_{t-1}$	0.409***	0.409***	0.363***
	(0.008)	(0.008)	(0.008)
$ln(Avg\ Wage)$	0.171***	0.171***	0.166***
	(0.004)	(0.004)	(0.005)
$ln(Avg\ RCA)$	0.014***	0.018***	0.019***
(Subsector-Level)	(0.002)	(0.002)	(0.002)
$ln(Avg\ RCA)_{t-1}$		-0.011***	-0.005**
(Subsector-Level)		(0.002)	(0.002)
$ln(Avg\ RCA)_{t-2}$			-0.003
(Subsector-Level)			(0.002)
Constant	1.306***	1.320***	1.438***
	(0.028)	(0.028)	(0.034)
Observations	786,305	786,305	636,260
R-squared	0.950	0.950	0.953
Year FE	Yes	Yes	Yes
Subsector x AMC FE	Yes	Yes	Yes
AMC x Year Clustered SE	Yes	Yes	Yes
Education Dummies	Yes	Yes	Yes

Notes: Robust standard errors in parentheses *** p<0.01, ** p<0.05, * p<0.1. Note that the comparable ln(Avg RCA) estimates in columns 1, 2, and 3, respectively, using the Arellano–Bond estimator, are .0427***, .056***, and .054*** respectively. The estimates of the first lag of ln(Avg RCA) in columns 2 and 3 are -.019*** and -.023*** respectively. The estimate of the second lag of ln(Avg RCA) is .003***

essentially means that the establishments being compared are those that had similar specialization levels during the previous year. The regression shown in column three includes both a one and two-year log RCA lag term. The coefficients in front of the one-year log RCA lag term in regressions (2) and (3) are both statistically significant but negative, which means the effect on growth from a higher RCA score is reduced when there is a control variable in place for an establishment's former level of labor specialization. The coefficient in front of the two-year log RCA lag term in regression (3) is not statistically significant which suggests only specialization levels for the previous year matter, not those from the years before that.

To be numerically precise, when the previous year's RCA level is added as a control variable, comparing two establishments that were of equal size (in the former year) shows that a one percent rise in RCA score is associated with a 0.007 percent rise, on average, in establishment size, instead of the 0.014 percent rise that would result when former specialization levels are uncontrolled (as in column 1). The implication is that some of the benefit to growth experienced by specialized establishments can be explained by former levels of specialization, with less specialized establishments experiencing more gains than more specialized establishments. When the initial level of labor specialization is uncontrolled, positive changes in labor specialization are associated with higher growth, but that growth is disproportionately caused by establishments that are initially less specialized.

The regression shown in the third column is the same as the regression shown in the second column, but it includes a two-year log RCA lag term as well. As mentioned, the coefficient on the two-year log RCA lag term is not statistically significant. The results in column three indicate that labor specialization has a slightly higher effect size on establishment labor pool size than that implied by regression (2). However, the number of observations are now smaller since two lags are used instead of one, which means there is slightly less variation in the data. In general, the effect sizes associated with each of the three regressions remain modest regardless of the number of lag terms employed.

It should be noted that the ARCA measure is more appropriate for comparisons across time than the RCA measure. Therefore, the regression results that utilize RCA as the complementarity measure should be treated as less reliable than the regression results that utilize ARCA. With that being said, although the results are not as reliable as the regression results that utilize the ARCA measure, the results shown in Table 3 can serve as a valuable basis with which to compare the regressions that utilize ARCA as the complementarity measure (which are shown in Table 4).

The results in table 4 are displayed in a manner identical to those in Table 3. Importantly, with the exception that ARCA is used as the complementarity score rather than RCA, the regression specification is the same, including the number of lag terms used in each regression. As can be seen, the effect of a one percent change in average ARCA on establishment size is now higher. Notice also that the R-squared values are about the same. Changing the complementarity score from RCA to ARCA leads to a higher effect size with no reduction in predictive power. The effect sizes associated with log establishment size and log average wages are the same as those in Table 3. The additional variation now explained by ARCA, when compared to RCA, is all the more impressive in light of these facts.

The results of the first regression in Table 4 shows that a one percent rise in the average establishment ARCA level is associated with a 0.036 percent rise in average establishment size. This is between two to three times as large as the effect size shown in Model 1 of Table 3. One may contend that the effect size, although higher than the one in Model 1 of Table 3, is still low overall. However, it is necessary to interpret the effect size in relation to the fact that average wages are held fixed. To be more precise, a one percent rise in the level of specialization, as measured by the ARCA index, is coming at no expense (in the form of wage-based labor cost) to the establishment since average wage is held fixed. Taken in this light, a 0.036 percent rise in establishment size is not a negligible amount.

Columns two and three of Table 4 show a positive effect size as lags are introduced. When one lag is introduced, the effect of log ARCA on log size is diminished slightly when comparing establishments with the same ARCA level for year t-1 (it goes from 0.036 to 0.027). Nevertheless, it remains noticeably higher than the effect size of log RCA on log size shown in column two of

Table 4: Regression of Log Establishment Size on Log ARCA

	Dependent Variable		
	$ln(Avg\ Size)$	$ln(Avg\ Size)$	$ln(Avg\ Size)$
Model	(1)	(2)	(3)
$ln(Avg\ Size)_{t-1}$	0.409***	0.409***	0.363***
	(0.008)	(0.008)	(0.008)
$ln(Avg\ Wage)$	0.172***	0.172***	0.167***
	(0.004)	(0.004)	(0.005)
$ln(Avg\ ARCA)$	0.036***	0.043***	0.034***
(Subsector-Level)	(0.008)	(0.009)	(0.010)
$ln(Avg\ ARCA)_{t-1}$		-0.016**	-0.003
(Subsector-Level)		(0.007)	(0.007)
$ln(Avg\ ARCA)_{t-2}$			-0.007
(Subsector-Level)			(0.006)
Constant	1.416***	1.398***	1.516***
	(0.026)	(0.026)	(0.032)
Observations	786,305	786,305	636,260
R-squared	0.950	0.950	0.952
Year FE	Yes	Yes	Yes
Subsector x AMC FE	Yes	Yes	Yes
AMC x Year Clustered SE	Yes	Yes	Yes
Education Dummies	Yes	Yes	Yes

Notes: Robust standard errors in parentheses *** p<0.01, ** p<0.05, * p<0.1. Note that the comparable ln(Avg ARCA) estimates in columns 1, 2, and 3, respectively, using the Arellano–Bond estimator, are -.0046, .0477***, and .042*** respectively. The estimates of the first lag of ln(Avg ARCA) in columns 2 and 3 are -.0726*** and -.0729*** respectively. The estimate of the second lag of ln(Avg ARCA) is .0001

Table 3. Finally, when two lags are introduced, the two lags lose their significance, in which case the effect size of log ARCA of log size resembles, with only a slight diminution, the value it has in column one.

Since the ARCA index is shown by Hoen & Oosterhaven (2006) to be a more appropriate measure for comparisons across time, it is justifiable on both empirical and theoretical grounds to consider the results in Table 4, as opposed to those in Table 3, as the more accurate of the two in estimating the effect of labor force specialization on establishment size. As a result, from hereon in, the index that will be used in upcoming regressions is the ARCA index.

Although the regression in Table 4 is illuminating, a more sophisticated isolation of the desired effect size would come from using an instrumental variable that only affects establishment growth through its effect on establishment specialization. The study here does not utilize instrumental variables, so endogeneity can not definitively be ruled out. One argument for a non-causal interpretation is that higher productivity establishments may become more productive through other means first. They then become large. Once they become large, they are able to professionalize their hiring practices. This professionalization then might lead to a greater ability to prune their workforce of redundant occupational tasks and to better optimize their hiring practices so that they hire occupations that are more relevant to the production of the goods and services which generate the most money for them.

Even in this case, however, the only reason they have to optimize their hiring practices in a manner that leads to labor force specialization is if there is an increase in their productivity associated with such specialization. The increase in productivity sought by them would need to be manifest in more spending on labor and capital. In theory, this need not entail an increase in the size of the labor force, but, for many establishments, it will entail exactly that. For larger establishments intent on growing, they may be able to specialize their labor force with greater ease than in comparable smaller establishments, but the incentive for specializing would only be there in the first place if it encourages future growth. At the very least, it would need to stave off the diminished growth that will inevitably come as they become too large. Nevertheless, further research is necessary to assess the direction of causality before any conclusive statements can be made in this area.

The regression shown in the first column of Table 4 is repeated in the first column of Table 5. The same regression is repeated so that the results can be compared to the results in the second and third columns of Table 5, which each employ a different combination of fixed-effects. Different choices of fixed effects help control for unobservables that subsector × AMC fixed effects may not sufficiently capture by themselves. For example, subsector × AMC fixed effects may not be sufficient to capture the effect that tariff changes have on establishment size. For that, there is a time-subsector interaction component that must be taken into account. Tariff changes in the 1990-1995 period in Brazil were industry-specific and gradually introduced (Kume, 1990; Kume et al., 2003). In order to control for the industry-time component on establishment growth caused by tariffs, subsector × year fixed effects need to be used.

Another example of an effect on establishment size that subsector × AMC fixed effects may not be sufficient to capture is that which is caused by the spatial component of a trade shock's effect (Caliendo et al., 2019), and, more specifically, the role that the structure of the input-output network across space has in mediating such an effect (Pierce & Schott, 2016; Acemoglu et al., 2016; Amiti & Cameron, 2004). An effect such as that may be operating as part of Brazil's trade shock. Due to the uneven geographic distribution of input-output links within and across industries, establishments in close geographic proximity to establishments in heavily affected industries are likely to be more heavily affected themselves as a result of that geographic proximity. In order to control for the effects in question, effects which are both localized due to the geographic component discussed and temporal due to how Brazil's trade liberalization was phased in, AMC × year fixed effects need to be used.

The fixed effects employed are as follows: in the second column of Table 5, subsector × year fixed effects are employed rather than subsector × AMC fixed effects and non-interacted AMC fixed effects are employed separately. Similarly, in (3), non-interacted subsector fixed effects are employed along with AMC × year fixed effects. The standard errors remain clustered on the AMC

Table 5: Regression of Log Establishment Size on Log ARCA (Different FE Choices)

	Dependent Variable		
	$ln(Avg\ Size)$	$ln(Avg\ Size)$	$ln(Avg\ Size)$
Model	(1)	(2)	(3)
$ln(Avg\ Size)_{t-1}$	0.409***	0.858***	0.855***
	(0.008)	(0.002)	(0.002)
$ln(Avg\ Wage)$	0.172***	0.113***	0.120***
	(0.004)	(0.003)	(0.004)
$ln(Avg\ ARCA)$	0.036***	0.044***	0.036***
(Subsector-Level)	(0.008)	(0.003)	(0.003)
Constant	1.416***	0.449***	0.420***
	(0.026)	(0.018)	(0.020)
Observations	786,305	802,687	801,160
R-squared	0.950	0.928	0.926
Subsector FE	No	No	Yes
AMC FE	No	Yes	No
Year FE	Yes	No	No
Subsector x Year FE	No	Yes	No
AMC x Year FE	No	No	Yes
Subsector x AMC FE	Yes	No	No
AMC x Year Clustered SE	Yes	Yes	Yes
Education Dummies	Yes	Yes	Yes

Notes: Robust standard errors in parentheses *** p<0.01, ** p<0.05, * p<0.1

× year level for each of the regressions. The estimated effect of log ARCA on log size remains the same in (3) and is only slightly larger in (2). The results give further support to the results found in Table 4. Notice that the effect size on the log size lag term changes quite drastically. Meanwhile, the estimated effect sizes of log ARCA barely change. The fact that the estimates are robust not only in their sign, but also in their magnitude, to various specifications, each with a different set of fixed effects, and that the same can not be said about the effect size of other variables like the establishment size lag term, bolsters the credibility of the log ARCA effect size estimate.

6.2 Relationship between Specialization and Establishment Size

The next topic of discussion is whether large establishments benefit more from specialization or not. It seems likely that larger establishments would benefit more since their larger size allows them to more effectively match individual workers to roles that are a better utilization of their skills

(Minni, 2024). Taking this into account, if an interaction between the log establishment size lag term and log ARCA is included in the regression, it should be positive. However, as discussed previously, there is also the possibility that a positive estimated value may be indicative of larger establishments having greater ease in specializing their workforce once they have attained a large size since, at such a point, they might already be on a trajectory of growth due to factors unrelated to their specialization. The factors causing their higher growth may also play a role at facilitating labor force specialization.

Table 6 includes a regression with the same specification as that which is shown in column one of Table 5 (and Table 4) with one change: the addition of a log size lag term by log ARCA interaction term. Additionally, Table 6 includes the results of two regressions, both of which have the same regression specification as in column one, but applied on two separate samples, one on nontradable subsectors (shown in column two) and one on tradable subsectors (shown in column three). The next few tables will, similarly, involve looking at regression results separately on those same two subsamples: tradable subsectors and nontradable subsectors. It should be noted that every subsector belongs to one, and only one, of the aforementioned categories.

The results in column one of Table 6 provide support for the hypothesis that large firms are better able to exploit the growth-enhancing benefits of labor force participation. When the interaction term is included in the regression, the non-interacted effect of the ARCA index on establishment size is negative. The median establishment has about seven full-time workers. Therefore, for the establishment of median size, a one percent rise in the ARCA score is associated with a $-0.065 + ln(7) * 0.055 \approx 0.042$ percent rise in establishment size.

For larger establishments, the effect is even greater. For an establishment in the 95th percentile of the size distribution, with 55 employees, the effect would be a $-0.065 + ln(55) * 0.055 \approx 0.155$ percent rise in establishment size. This effect is striking when one considers that establishments experience diminished growth rates as they get larger. Tentatively, it can be said that one way for an establishment to stave off the diminution in growth that comes with attaining a larger size is to become more specialized. Whether such an assertion is fully reliable or not depends on whether the estimated effect remains biased upward by the reverse causality of the form mentioned earlier: that which emanates from establishments with higher growth rates having greater ease in specializing their workforce. Placing a log size lag term may attenuate some of that bias but not all of it.

Columns two and three show that the effect size on the interaction between average establishment size and the average establishment-level ARCA score is much higher in the nontradable subsectors than in the tradable subsectors. The effect size of non-interacted log ARCA is negative and of greater magnitude in column 2. For tradables, the effect size of non-interacted log ARCA is not significant and the effect size of the interaction term is about a third of the size of the effect size shown in column two.

6.3 Comparisons across Productivity Measures

So far, the outcome variable of interest has been log size. Although higher productivity can lead to establishment growth, there are other channels through which higher productivity can be manifested. Unfortunately, RAIS data does not have establishment-level sales and revenue data. It also does not record establishment-level investment in capital. Nevertheless, there are two other productivity measures that can be used as the outcome variable: wages, and total wages. The wage variable, i.e. establishment level average wage, has already been used in previous regressions

Table 6: Log ARCA by Log Establishment Size Interaction Effect

	Dependent Variable		
	$ln(Avg\ Size)$	$ln(Avg\ Size)$	$ln(Avg\ Size)$
Model	(1)	(2)	(3)
$ln(Avg\ Size)_{t-1}$	0.536***	0.561***	0.431***
	(0.017)	(0.021)	(0.017)
$ln(Avg\ ARCA)$	-0.065***	-0.107***	0.021
(Subsector-Level)	(0.008)	(0.009)	(0.015)
$ln(Avg\ Wage)$	0.172***	0.159***	0.220***
	(0.004)	(0.005)	(0.007)
$ln(Avg\ ARCA)$ x $ln(Avg\ Size)_{t-1}$	0.055***	0.064***	0.019***
	(0.006)	(0.007)	(0.007)
Constant	1.178***	1.040***	1.612***
	(0.035)	(0.042)	(0.047)
Observations	786,305	520,631	265,674
R-squared	0.950	0.952	0.937
Year FE	Yes	Yes	Yes
Subsector x AMC FE	Yes	Yes	Yes
AMC x Year Clustered SE	Yes	Yes	Yes
Education Dummies	Yes	Yes	Yes

Notes: Robust standard errors in parentheses *** p<0.01, ** p<0.05, * p<0.1. If the Arello-Band estimator is used, the column 1 ln(Avg ARCA) estimate becomes 0.220***, and the estimate of $ln(Avg\ ARCA)$ x $ln(Avg\ Size)_{t-1}$ becomes -.71***.

as an independent variable. One way of exploring whether the positive effect size of labor force specialization on establishment size is due to higher productivity or not is by comparing the results of Tables 4 and 5 to the results of the same regressions where the dependent variable is replaced by log wage (or log total wages). "Total wages" is defined as the product of the average wage and the number of employees. Therefore, it captures the proceeds of higher productivity that might go not only into more hiring but also into higher wages paid to employees.

Table 7 has the results of three regressions. The first column contains the results of the same regression shown in column one of Tables 5 and 4. No changes were made to it. It is included in Table 7 so that it can be compared to the results in columns two and three. The regression in column two is similar to the regression in column one, with the sole difference being that log wage replaces log size as the dependent variable, a log wage lag term is included rather than a log size lag term as an independent variable, and log size is included as an independent variable rather than log wage.

In column three, log size and log wage are removed from the regression entirely since size and wage multiplied together becomes total wages. Instead, log total wages is the dependent variable and the independent variables are the log total wages lag term and log ARCA. As is typical, subsector by year fixed effects are employed, clustered standard errors are calculated on the level of AMC by year, and education dummies are included in the regression.

Table 7: Comparison of ARCA's Effect on Three Different Dependent Variables

	Dependent Variable		
	$ln(Avg\ Size)$	$ln(Avg\ Wage)$	$ln(Total\ Wage)$
Model	(1)	(2)	(3)
$ln(Avg\ Size)_{t-1}$	0.409***		
	(0.008)		
$ln(Avg\ Wage)$	0.172***		
	(0.004)		
$ln(Avg\ Size)$		0.102***	
		(0.002)	
$ln(Avg\ Wage)_{t-1}$		0.237***	
		(0.006)	
$ln(Total\ Wage)_{t-1}$			0.396***
			(0.008)
$ln(Avg\ ARCA)$	0.036***	-0.087***	-0.054***
(Subsector-Level)	(0.008)	(0.004)	(0.009)
Constant	1.416***	-0.175***	1.482***
	(0.026)	(0.018)	(0.033)
Observations	786,305	783,629	783,629
R-squared	0.950	0.928	0.956
Year FE	Yes	Yes	Yes
Subsector x AMC FE	Yes	Yes	Yes
AMC x Year Clustered SE	Yes	Yes	Yes
Education Dummies	Yes	Yes	Yes

Notes: Robust standard errors in parentheses *** p<0.01, ** p<0.05, * p<0.1

The results in Table 7 show that the effect size of log ARCA on both log wages and log total wages is negative and significant. Moreover, the magnitude of each effect size is larger than the magnitude of the effect size of log ARCA on log size. Taken on their own, these results suggest

serious reconsideration of whether specialization, expressed through a higher ARCA score, is actually productivity enhancing or not. These results alone do not definitively provide the answer though.

Unless the full general equilibrium effects of specialization on employees hired (and on market wages) are known, the situation where the productivity-enhancing benefits of specialization is entirely captured through more hiring alone (and not higher wages) can not be ruled out. After all, the model shown in 2.2 provides a framework for how specialization can be productivity enhancing even though those benefits do not get capitalized into higher wages, and, in fact, do not even benefit individual establishments. Instead, they influence the specialization patterns writ large and cause higher-than-average hiring of occupations by subsectors to which their occupational skills are more complementary.

The results shown in Tables 3 to 5 support the notion that subsectors in AMCs that hire more occupations that are complementary to their subsector and fewer that are not will experience higher levels of employee growth in the future. The findings do not definitively show that these higher levels of growth stem, causally, from hiring more complementary occupations. Nevertheless, the relationship still stands, and it is important to understand that the relationship can happen, i.e., higher labor force specialization and productivity, with no changes to the average wages of the occupations hired.

To further shed light on the matter, consider that if an establishment in a certain subsector-AMC pair experiences higher future employee growth compared to an establishment in another subsector-AMC pair, that does not mean that these establishments have a higher profit or premium on their goods over those with less future employee growth. Similarly, it is not the case that establishments within a market that experiences higher future employee growth are making a higher profit than establishments within the same market with lower employee growth. As discussed in Subsection 3.2, applying the Eaton & Kortum (2002) framework to a labor market setting where establishments belonging to specific subsectors compete for occupations both within and across subsectors ensures the specialization can happen within a market that is competitive.

The model framework in Subsection 3.2 translates higher market-decided levels of sales, that competitive establishments within a market must meet, into more hiring. Employees, being price takers, will be hired in greater numbers but their wages will remain the same. The T_s^o parameter across subsectors and occupations governs the ratio of employees who will be hired in one occupation over another for a given subsector. Employees are infinitely flexible in what occupations they pursue and they all work for their reservation wage since there is sufficient competition provided by unemployed people who can reenter the workforce. It is these latter assumptions on employee behavior that prevents higher productivity within a subsector from leading to higher wages within that subsector, within the context of the model. Whether the relationship suggested by the model can explain the results Table 7 depends on how much market power occupations have. The main point is that the lack of a positive relationship between the ARCA index and average wages does not, in and of itself, imply that labor force specialization, measured by the ARCA index, is not productivity enhancing.

One counter-argument that can be made is that the relationship in Table 7 between ARCA and average wage is negative. The lack of a positive relationship can be explained by the model proposed in Subsection 3.2, but not a negative relationship. This is a valid point, and one needs to step outside the model framework proposed in Subsection 3.2 to look for a valid explanation. Nevertheless, there are several possible forces that may be operating which can explain the results in columns 2 and 3,

although more research is required to know for certain which forces are present and which ones aren't.

One possibility is that there is a sufficiently high enough relationship between firm labor force specialization and market power in labor force hiring to give firms (and the establishments they own) more market power in setting wages, which then grants these firms some ability in lowering wages from the perfectly competitive scenario. In this case, it is presumed that perfectly competitive wages are above the reservation wage. Testing this hypothesis requires access to the identified data to know which firms individual establishments belong to.

It is important to keep in mind that because subsector by AMC fixed effects are employed, only within subsector-AMC variation is being used to come up with the estimates. In effect, the goal is to isolate shocks to labor force specialization that could lead to higher productivity within the establishment that could then lead to the establishment raising wages to attract a more adept, more skilled, work force. Although that is the goal, without employing an instrumental variable approach, one can never be sure that the variation in labor force specialization, even after employing fixed effects, comes from shocks that wages are not also independently being exposed to.

It is even possible that average wage itself is affecting labor force specialization, in which case reverse causality is at work. To demonstrate this point, consider a technology shock that causes a notable automation of tasks for a certain occupation. This shock might make training for said occupation cheaper and might open it up to a larger number of people, thus lowering the wages. Concomitantly, it might have the effect of encouraging labor force specialization among the establishments that hire said occupation. One example is found in the publishing industry after the invention of desktop publishing (Bessen, 2015). The invention of desktop publishing made it vastly easier for someone with little formal training to enter the industry. This had the effect of lowering wages. It also led to firms outsourcing their publishing tasks to freelancers and small specialized firms. Research suggests that firms respond to an increase in competition by becoming more specialized (Backus, 2020).

Although Bessen (2015) makes clear that, in the long run, workers who are better at adapting can raise their wages back up, this process can take years. Connecting this research back to the results in Table 7, it is possible that there are technological trends causing average wages to lower for certain occupations, and, subsectors which mainly employ those occupations, respond by specializing. Their response may be directly motivated by the lower wages of the occupations they mainly hire or it can be part of a larger process of technological change that leads to subsector wide reorganization. It is the latter possibility that will be explored as it pertains to the question of new work. No definitive conclusions will be offered but suggestive evidence indicating the presence of new work will be shown in **??**.

Before proceeding to the topic of new work, a further look into the relationship between ARCA and the three productivity measures, i.e., average size, average wage, and total wages, is in order. Tables 8,9 and 10, deal with each of these productivity measures separately (Table 8 with average size, Table 9 with average wage, and Table 10 with total wages). Furthermore, now the relationship between density, ARCA, and the productivity measures can be given fuller treatment.

In light of the motivating evidence discussed in Tables 1-3, it is worth inquiring whether there may be an interaction effect between ARCA and density. Tables 8-10 include log density as an independent variable and the interaction between log density and log ARCA as another independent variable. The three regressions associated with each table correspond to different sample size restrictions. The first regression in each table is the regression on the full sample. The second

and third regression in each table are on nontradable sectors and tradable sectors, respectively. Separating the sample by whether the subsector is tradable or nontradable is relevant in so far as tradable/nontradable sectors derive different levels of benefit from labor force specialization. If so, highlighting this dynamic is useful in its own right since it allows for a more detailed understanding of the specific subsectors of the economy that are responsible for the relationship. In addition to that, it provides information helpful to the understanding the nature of ARCA's effect on wages. If it happens to be the case that technological trends may be responsible for shocks that reduce wages but raise labor force specialization concomitantly, there is the question of whether this might be less the case in the nontradable sector than the tradable sector.

It should be noted that the nontradable sector, although less reliant on capital, is not immune to shocks in technology affecting its labor organization. Even so, the tradable sector also is likely to have subsectors exposed to fiercer competition (Melitz & Ottaviano, 2008), which means they may be more responsive to the kinds of technological shocks described earlier that may simultaneously lower wages for the employees they hire and incentivize specialization on their part.

Although Tables 8-10 involve regressions on different cross-sections of the sample, the observations remain at the subsector by AMC level, the standard errors remain clustered at the AMC by year level, and the regressions continue to be weighted by establishment count. Some striking trends make themselves known with the new regression specifications, as can be seen in Table 8 below.

Once density levels are controlled and a log density-log ARCA interaction term is included, the first regression in Table 8 shows how much larger the effect size of log ARCA, one that doubles for establishments in AMC geographies of median density. Assuming median density levels (i.e. .0000798 people per square feet for the AMC, which corresponds to about 206 people per square mile), a 1% rise in the ARCA index leads to a $0.321 + 0.026 * ln(.0000798) \approx .08\%$ rise in average establishment size. When comparing observations at the 75th percentile AMC density levels, a 1% rise in the ARCA index leads to about a .15% rise in average establishment size and the comparable figure for observations in the 95th AMC density levels is about a .16% rise.

For the nontradable sector (the second column), assuming median density levels (i.e. .0000798 people per square feet for the AMC, which corresponds to about 206 people per square mile), a 1% rise in the ARCA index leads to a $0.181 + 0.016 * ln(.0000798) \approx .03\%$ rise in average establishment size. When comparing observations at the 75th percentile AMC density levels, a 1% rise in the ARCA index leads to about a .07% rise in average establishment size and the comparable figure for observations in the 95th AMC density levels is about a .08% rise.

For the tradable sector (the third column), assuming median density levels (i.e. .0000798 people per square feet for the AMC, which corresponds to about 206 people per square mile), a 1% rise in the ARCA index leads to a $0.440 + 0.035 * ln(.0000798) \approx .11\%$ rise in average establishment size. When comparing observations at the 75th percentile AMC density levels, a 1% rise in the ARCA index leads to about a .2% rise in average establishment size and the comparable figure for observations in the 95th AMC density levels is about a .22% rise.

Clearly, there is a large difference in the effect size of ARCA in the tradable sector compared to the nontradable sector. The results suggest that tradable sector establishments benefit more from labor force specialization than nontradable sector establishments do. These results are a striking contrast from the ones shown in Table 9, where log average wage is the dependent variable.

The results in Table 9 show a negative relationship between labor force specialization and wages. Assuming median density levels (i.e. .0000798 people per square feet for the AMC,

Table 8: Log Size on Log ARCA (Tradable vs Nontradable)

	Dependent Variable		
	$ln(Avg\ Size)$	$ln(Avg\ Size)$	$ln(Avg\ Size)$
Sample	Full Sample	Nontradable	Tradable
Model	(1)	(2)	(3)
$ln(Avg\ Size)_{t-1}$	0.414***	0.419***	0.388***
	(0.008)	(0.010)	(0.007)
$ln(Avg\ Wage)$	0.179***	0.166***	0.227***
	(0.005)	(0.006)	(0.008)
$ln(Avg\ ARCA)$	0.321***	0.181***	0.440***
(Subsector-Level)	(0.055)	(0.065)	(0.053)
$ln(density)$	0.195***	0.166***	0.277***
	(0.011)	(0.012)	(0.021)
$ln(density)$ x $ln(Avg\ ARCA)$	0.026***	0.016***	0.035***
	(0.004)	(0.005)	(0.004)
Constant	3.335***	2.905***	4.460***
	(0.122)	(0.141)	(0.221)
Observations	672,067	433,661	238,406
R-squared	0.951	0.953	0.939
Year FE	Yes	Yes	Yes
Subsector x AMC FE	Yes	Yes	Yes
AMC x Year Clustered SE	Yes	Yes	Yes
Education Dummies	Yes	Yes	Yes
Full Sample	Yes	No	No
Nontradable	No	Yes	No
Tradable	No	No	Yes

Notes: Robust standard errors in parentheses *** p<0.01, ** p<0.05, * p<0.1

which corresponds to about 206 people per square mile), a 1% rise in the ARCA index leads to a $-0.219 + -0.012 * ln(.0000798) \approx -.11\%$ reduction in average establishment size. When comparing observations at the 75th percentile AMC density levels, a 1% rise in the ARCA index leads to about a -.14% reduction in average establishment size and the comparable figure for observations in the 95th AMC density levels is about a -.14% reduction.

For the nontradable sector, assuming median density levels (i.e. .0000798 people per square feet for the AMC, which corresponds to about 206 people per square mile), a 1% rise in the ARCA

Table 9: Log Wage on Log ARCA (Tradable vs Nontradable)

	Dependent Variable		
	$ln(Avg\ Wage)$	$ln(Avg\ Wage)$	$ln(Avg\ Wage)$
Sample	Full Sample	Nontradable	Tradable
Model	(1)	(2)	(3)
$ln(Avg\ Size)$	0.107***	0.117***	0.092***
	(0.002)	(0.003)	(0.003)
$ln(Avg\ Wage)_{t-1}$	0.237***	0.257***	0.168***
	(0.007)	(0.008)	(0.007)
$ln(Avg\ ARCA)$	-0.219***	-0.207***	-0.206***
(Subsector-Level)	(0.030)	(0.042)	(0.036)
$ln(density)$	-0.053***	-0.042***	-0.097***
	(0.013)	(0.013)	(0.018)
$ln(density)$ x $ln(Avg\ ARCA)$	-0.012***	-0.010***	-0.013***
	(0.002)	(0.003)	(0.003)
Constant	-0.739***	-0.653***	-1.076***
	(0.125)	(0.132)	(0.183)
Observations	669,991	432,437	237,554
R-squared	0.930	0.930	0.931
Year FE	Yes	Yes	Yes
Subsector x AMC FE	Yes	Yes	Yes
AMC x Year Clustered SE	Yes	Yes	Yes
Education Dummies	Yes	Yes	Yes
Full Sample	Yes	No	No
Nontradable	No	Yes	No
Tradable	No	No	Yes

Notes: Robust standard errors in parentheses *** p<0.01, ** p<0.05, * p<0.1

index leads to a $-0.207 + -0.010 * ln(.0000798) \approx -.11\%$ reduction in average establishment size. When comparing observations at the 75th percentile AMC density levels, a 1% rise in the ARCA index leads to about a -.14% reduction in average establishment size and the comparable figure for observations in the 95th AMC density levels is about a -.14% reduction.

For the tradable sector, assuming median density levels (i.e. .0000798 people per square feet for the AMC, which corresponds to about 206 people per square mile), a 1% rise in the ARCA index leads to a $-0.206 + -0.013 * ln(.0000798) \approx -.08\%$ reduction in average establishment size.

When comparing observations at the 75th percentile AMC density levels, a 1% rise in the ARCA index leads to about a -.12% reduction in average establishment size and the comparable figure for observations in the 95th AMC density levels is about a -.12% reduction.

On top of the results showing a negative relationship between labor force specialization and wages, it also differs from the results in Table 8 because there is no noteworthy difference between the tradable sector results and the nontradable sector results. The effect sizes are negative but of equal magnitude between the tradable and nontradable sectors. So far, the results can be summarized as follows: in so far as labor force specialization has a positive effect on an establishment's productivity, this effect leads to a large pool of workers but not higher wages, and it is present in the tradable sector to a greater degree than the nontradable sector.

One question that arises is whether one should consider the results in Table 9 as suggestive evidence for a decline in productivity. If this were the case, then the decline in productivity implied by Table 9 needs to be balanced with the increase in productivity implied by Table 8. Table 10, below, shows the results for the same three regressions with the dependent variable now being the log of total wages. Since total wages is calculated by multiplying the labor force pool times average wages, a regression with log total wages as the dependent variable should include both the productivity enhancing role of labor force specialization and productivity reducing role of labor force specialization in the estimated effect size of the ARCA index. What do the results show?

The results in Table 10 indicate a positive effect size in the tradable sector and a negative effect size in the nontradable sector. Assuming median density levels (i.e. .0000798 people per square feet for the AMC, which corresponds to about 206 people per square mile), a 1% rise in the ARCA index leads to a $0.136 + 0.018 * ln(.0000798) \approx -.03\%$ rise in average establishment size. When comparing observations at the 75th percentile AMC density levels, a 1% rise in the ARCA index leads to about a .01% rise in average establishment size and the comparable figure for observations in the 95th AMC density levels is about a .02% rise.

For the nontradable sector, assuming median density levels (i.e. .0000798 people per square feet for the AMC, which corresponds to about 206 people per square mile), a 1% rise in the ARCA index leads to a $-0.014 + 0.008 * ln(.0000798) \approx -.09\%$ rise in average establishment size. When comparing observations at the 75th percentile AMC density levels, a 1% rise in the ARCA index leads to about a -.07% rise in average establishment size and the comparable figure for observations in the 95th AMC density levels is about a -.06% rise.

For the tradable sector, assuming median density levels (i.e. .0000798 people per square feet for the AMC, which corresponds to about 206 people per square mile), a 1% rise in the ARCA index leads to a $0.284 + 0.027 * ln(.0000798) \approx .03\%$ rise in average establishment size. When comparing observations at the 75th percentile AMC density levels, a 1% rise in the ARCA index leads to about a .1% rise in average establishment size and the comparable figure for observations in the 95th AMC density levels is about a .11% rise.

Taken together, these results suggest that, even if a negative relationship between labor force participation and wages is to be treated as productivity reducing, labor force specialization continues to be productivity enhancing in the tradable sector. Whereas in the nontradable sector, the reduction in wages is not balanced by an increase in labor force size, in the tradable sector a reduction in wages is balanced by an increase in labor force size such that the effect size of labor force specialization on total wages ends up being positive.

The different results when comparing the tradable sector to the nontradable sector is too large to ignore. Future work can identify the source of this discrepancy and research its relationship to

Table 10: Log Total Wage on Log ARCA (Tradable vs Nontradable)

	Dependent Variable		
	$ln(Tot\ Wage)$	$ln(Tot\ Wage)$	$ln(Tot\ Wage)$
Sample	Full Sample	Nontradable	Tradable
Model	(1)	(2)	(3)
$ln(Tot\ Wage)_{t-1}$	0.402***	0.411***	0.372***
	(0.001)	(0.001)	(0.002)
$ln(Avg\ ARCA)$	0.136***	-0.014	0.284***
(Subsector-Level)	(0.015)	(0.018)	(0.028)
$ln(density)$	0.179***	0.158***	0.227***
	(0.006)	(0.007)	(0.012)
$ln(density)$ x $ln(Avg\ ARCA)$	0.018***	0.008***	0.027***
	(0.001)	(0.002)	(0.002)
Constant	3.208***	2.823***	4.110***
	(0.062)	(0.073)	(0.122)
Observations	669,991	432,437	237,554
R-squared	0.958	0.959	0.953
Year FE	Yes	Yes	Yes
Subsector x AMC FE	Yes	Yes	Yes
AMC x Year Clustered SE	Yes	Yes	Yes
Education Dummies	Yes	Yes	Yes
Full Sample	Yes	No	No
Nontradable	No	Yes	No
Tradable	No	No	Yes

Notes: Robust standard errors in parentheses *** p<0.01, ** p<0.05, * p<0.1

agglomeration economies.

7 Conclusion

This thesis has examined how subsector–occupation complementarity—a measure adapted from the revealed comparative advantage framework—relates to establishment-level growth in Brazil's formal labor market during 1985–1993. The empirical analysis shows that greater alignment between an establishment's occupational composition and its subsector's comparative advantage is associated with larger establishment size, even after controlling for wages, educational composition, and fixed

effects. The results are robust across alternative specifications and measurement approaches.

A key finding is that the benefits of specialization are not evenly distributed: larger establishments appear better able to convert specialized labor into growth-enhancing outcomes, suggesting managerial capacity and scale effects play an important role. Furthermore, sectoral differences matter. In tradable subsectors, specialization's positive effects on establishment size and total wages are strong enough to offset wage reductions, whereas in nontradable subsectors these gains are weaker and less consistent. This divergence points toward structural and competitive differences that may shape how localization and urbanization economies translate into productivity.

Taken together, the evidence offers suggestive support for the view that the productivity benefits of specialization in this context are more closely linked to localization economies—those arising from industry-specific clustering—than to urbanization economies based on cross-industry diversity. While this interpretation is consistent with patterns in the data, establishing a definitive causal channel between specialization and specific types of agglomeration economies will require further research, ideally with microdata tracking establishment dynamics and worker flows over time.

Ultimately, the analysis underscores the importance of considering the fit between occupational structures and sectoral environments when evaluating establishment performance. As labor markets evolve and new forms of work emerge, understanding how such complementarities operate—and how they interact with spatial economic forces—remains a fertile avenue for future inquiry.

References

Accetturo, Antonio, Valter Di Giacinto, Giacinto Micucci, and Marcello Pagnini. 2018. "Geography, productivity and trade: Does selection explain why some locations are more productive than others?" *Journal of Regional Science* 58(5): 949–979.

Acemoglu, Daron, Ufuk Akcigit, and William R. Kerr. 2016. "Networks and the Macroeconomy: An Empirical Exploration." In NBER Macroeconomics Annual 2015, edited by Martin Eichenbaum and Jonathan A. Parker, 273–335. Chicago, IL: University of Chicago Press.

Amiti, Mary, and Lisa Ann Cameron. 2004. "Economic Geography and Wages: The Case of Indonesia." IMF Working Paper No. 2004/079. Washington, DC: International Monetary Fund.

Arrow, Kenneth J. 1962. "The Economic Implications of Learning by Doing." *The Review of Economic Studies* 29(3): 155–173. :contentReference[oaicite:2]index=2

Backus, Matthew. 2020. "Why Is Productivity Correlated with Competition?" *Econometrica* 88(6): 2415–2444.

Balassa, Bela. 1965. "Trade Liberalization and 'Revealed' Comparative Advantage." *The Manchester School of Economic and Social Studies* 33: 99–123.

Berg, Janine. 2010. *Laws or Luck? Understanding Rising Formality in Brazil in the 2000s.* Brasília: International Labour Office.

Bernhofen, Daniel M. 2005. "Gottfried Haberler's 1930 Reformulation of Comparative Advantage in Retrospect." *Review of International Economics* 13(5): 997–1000.

Bessen, James. 2015. *Learning by Doing: The Real Connection Between Innovation, Wages, and Wealth.* New Haven, CT: Yale University Press.

Caliendo, Lorenzo, Maximiliano Dvorkin, and Fernando Parro. 2019. "Trade and Labor Market Dynamics: General Equilibrium Analysis of the China Trade Shock." *Econometrica* 87(3): 741–835.

Combes, Pierre-Philippe, Gilles Duranton, Laurent Gobillon, Diego Puga, and Sébastien Roux. 2011. "The Identification of Agglomeration Economies." *Journal of Economic Geography* 11(2): 253–266.

Combes, Pierre-Philippe, Gilles Duranton, Laurent Gobillon, Diego Puga, and Sébastien Roux. 2012. "The Productivity Advantages of Large Cities: Distinguishing Agglomeration from Firm Selection." *Econometrica* 80(6): 2543–2594.

Del Gatto, Massimo, Adriana Di Liberto, and Carmelo Petraglia. 2011. "Measuring Productivity." *Journal of Economic Surveys* 25(5): 952–1008.

Dix-Carneiro, Rafael, and Brian K. Kovak. 2017. "Trade Liberalization and Regional Dynamics." *American Economic Review* 107(10): 2908–2946.

Duranton, Gilles, and Diego Puga. 2001. "Nursery Cities: Urban Diversity, Process Innovation, and the Life-Cycle of Products." *American Economic Review* 91(5): 1454–1477.

Duranton, Gilles, and Henry G. Overman. 2005. "Testing for Localization Using Micro-Geographic Data." *The Review of Economic Studies* 72(4): 1077–1106.

Duranton, Gilles, and Diego Puga. 2005. "From Sectoral to Functional Urban Specialisation." *Journal of Urban Economics* 57(2): 343–370.

Eaton, Jonathan, and Samuel Kortum. 2002. "Technology, Geography, and Trade." *Econometrica* 70(5): 1741–1779.

Eckert, Fabian, Sharat Ganapati, and Conor Walsh. 2020. "Skilled Scalable Services: The New Urban Bias in Economic Growth." CESifo Working Paper No. 8705. Munich, Germany: Center for Economic Studies and Ifo Institute (CESifo).

Eeckhout, Jan. 2021. *The Profit Paradox: How Thriving Firms Threaten the Future of Work.* Princeton, NJ: Princeton University Press.

Ehrl, Philipp. 2017. "Minimum Comparable Areas for the Period 1872–2010: An Aggregation of Brazilian Municipalities." *Estudos Econômicos (São Paulo)* 47(1): 215–229.

Faberman, R. Jason, and Matthew Freedman. 2016. "The urban density premium across establishments." *Journal of Urban Economics* 93(C): 71–84.

French, Scott. 2017. "Revealed Comparative Advantage: What Is It Good For?" *Journal of International Economics* 106: 83–103.

Gaubert, Cécile. 2018. “Firm Sorting and Agglomeration.” *American Economic Review* 108(11): 3117–3153.

Glaeser, Edward L., Hedi D. Kallal, José A. Scheinkman, and Andrei Shleifer. 1992. “Growth in Cities.” *Journal of Political Economy* 100(6): 1126–1152.

Goldschmidt, Deborah, and Johannes F. Schmieder. 2017. “The Rise of Domestic Outsourcing and the Evolution of the German Wage Structure.” *The Quarterly Journal of Economics* 132(3): 1165–1217.

Haberler, Gottfried. 1930. “Die Theorie der komparativen Kosten und ihre Auswertung für die Begründung des Freihandels.” *Weltwirtschaftliches Archiv* 32: 349–370.

Hayek, Friedrich A. 1988. *The Fatal Conceit: The Errors of Socialism*. Edited by W.W. Bartley III. Chicago: University of Chicago Press.

Henderson, J. Vernon. 1986. “Efficiency of Resource Usage and City Size.” *Journal of Urban Economics* 19(1): 47–70.

Hoen, Alex R., and Jan Oosterhaven. 2006. “On the measurement of comparative advantage.” *The Annals of Regional Science* 40(3): 677–691.

Jacobs, Jane. 1969. *The Economy of Cities.* New York: Vintage Books.

Jara-Figueroa, Christian, Spencer Yang Jun, Edward L. Glaeser, and César A. Hidalgo. 2018. “The Role of Industry-Specific, Occupation-Specific, and Location-Specific Knowledge in the Growth and Survival of New Firms.” *Proceedings of the National Academy of Sciences* 115(50): 12646–12653.

Kume, Honório. 1990. “A política tarifária brasileira no período 1980–88: avaliação e reforma.” Rio de Janeiro: INPES/IPEA (Série ÉPICO, 17).

Kume, Honório, Guida Piani, and Carlos F. Souza. 2003. “A política brasileira de importação no período 1987–98: descrição e avaliação.” In C. Corseuil and H. Kume (Eds.), *A abertura comercial brasileira nos anos 1990*. Rio de Janeiro: Ipea.

Lin, Jeffrey. 2011. “Technological Adaptation, Cities, and New Work.” *Review of Economics and Statistics* 93(2): 554–574.

Melitz, Marc J. 2003. “The Impact of Trade on Intra-Industry Reallocations and Aggregate Industry Productivity.” *Econometrica* 71(6): 1695–1725.

Melitz, Marc J., and Gianmarco I.P. Ottaviano. 2008. “Market Size, Trade, and Productivity.” *The Review of Economic Studies* 75(1): 295–316.

Metcalfe, Robert D., Alexandre B. Sollaci, and Chad Syverson. 2023. “Managers and Productivity in Retail.” NBER Working Paper No. 31192.

Minni, Virginia. 2024. “Making the Invisible Hand Visible: Managers and the Allocation of Workers to Jobs.” Working paper posted November 15, 2024. SSRN.

Neffke, Frank, and Martin Henning. 2013. "Skill relatedness and firm diversification." *Strategic Management Journal* 34(3): 297–316.

Li, Yang, and Frank Neffke. 2024. "Evaluating the Principle of Relatedness: Estimation, Drivers and Implications for Policy." *Research Policy* 53(3): 104952.

Pierce, Justin R., and Peter K. Schott. 2016. "The Surprisingly Swift Decline of US Manufacturing Employment." *American Economic Review* 106(7): 1632–1662.

Polanyi, Michael. 1958. *Personal Knowledge: Towards a Post-Critical Philosophy*. Chicago: University of Chicago Press.

Rosenthal, Stuart S., and William C. Strange. 2003. "Geography, Industrial Organization, and Agglomeration." *Review of Economics and Statistics* 85(2): 377–393.

Syverson, Chad. 2004. "Market Structure and Productivity: A Concrete Example." *Journal of Political Economy* 112(6): 1181–1222.

Ulyssea, Gabriel. 2018. "Firms, Informality, and Development: Theory and Evidence from Brazil." *American Economic Review* 108(8): 2015–2047.

Appendix

A Model

A.1 Labor Market Structure and Matching Process

This section elaborates on the two-stage labor hiring process described in the model. There are S subsectors and O occupations. Workers are fully indifferent to their initial choice of occupational training. Consequently, the distribution of workers across occupations reflects the full range of skill diversification types, denoted by $\zeta \in \Omega$. There exists an inexhaustible supply of workers; whenever a worker trained in occupation o and associated with a given ζ is hired, another identical worker immediately enters the labor pool.

To clarify the role of ζ, it represents a worker's broad approach to skill diversification. Formal training provides a foundation of occupational skills, but workers, in choosing their training path, randomly (in a uniformly distributed manner) select a specialization strategy — a ζ value representing a strategy lying on a generalist-to-specialist range. As workers are otherwise identical in learning ability, all strategies require equal effort. Once this choice is made, it crystallizes into fixed habits, meaning a worker's ζ value remains immutable across all occupations and subsectors. Importantly, this choice does not affect expected productivity in an ex ante sense, nor does it determine workers' skillsets; rather, it represents an approach to skill diversification. The particular skills a worker has access to comes from a combination of formal and tacit learning. The specific skills that get employed comes from an idea, made by the worker, on how best to employ them. Variation in realized productivity stems from how workers embody their skills for whatever chosen level of specialization they have, and, since the embodiment of skills stems from an idea, a Fréchet distribution is used to model productivity.

The market is one where workers with the best idea get hired. The process is as follows: workers, regardless of specialization degree, propose ideas for how their skills can be applied to the occupation-subsector setting associated with the position for which they are being interviewed. Being a specialist or generalist, in itself, is uninformative to how productive a worker will be without knowing which skills are emphasized and how they are deployed. For instance, a specialist programmer may master a single language, while a generalist may have competence across multiple languages. In either case, productivity depends on how these skills are applied to specific technological and organizational environments.

One may ask, what purpose does specialization hold in this model? Just like the Eaton and Kortum (2002) model uses the full spectrum of varieties across goods to create a model that allows for realistic trade flows across countries, this model utilizes the spectrum of specialization types to allow for realistic patterns of employment across subsectors. Poaching is rarely a one-way affair, even for a single occupation. Tech may poach data scientists from the finance industry even as the finance industry poaches data scientists from tech, and this is the pattern that would most likely be seen for most occupations. The exchange happens in both directions rather than in one, and this model allows for these realistic exchange patterns to result.

The hiring process unfolds in two stages:

Stage 1: Hiring First-Time Workers (Subsector-Occupation Autarky)

Each worker selects an occupation for training and draws a ζ value. Within each (s, o) pair,

there is positive mass of workers with every ζ type. Subsector establishments solve the following cost-minimization problem:

$$\min_{\{l_s^o(\zeta) \mid \forall s, \forall o, \zeta \in \Omega\}} c_s = \left(\sum_o \int_\Omega c_{s,s}^{o,o}(\zeta)\ l_s^o(\zeta)\ d\zeta \right) \tag{35}$$

subject to:

$$\left(\sum_o \tilde{\beta}_s^o (L_s^o)^{\frac{\sigma_s - 1}{\sigma_s}} \right)^{\frac{\sigma_s}{\sigma_s - 1}} \geq \overline{Q_s} \tag{36}$$

where $\overline{Q_s}$ is the exogenously determined quantity of output demanded from subsector s by consumers, σ_s is the elasticity of substitution across occupations, and $\tilde{\beta}_s^o$ is a demand shifter. Labor aggregation in subsector s across skill-diversification types for occupation o follows:

$$L_s^o = \left(\int_\Omega l_s^o(\zeta)^{\frac{\eta_s^o - 1}{\eta_s^o}} d\zeta \right)^{\frac{\eta_s^o}{\eta_s^o - 1}} \tag{37}$$

In this first stage, establishments only hire people for occupational roles for which they received training. They are not able to poach from other subsectors since no one has been hired before stage 1. The $c_{s,s}^{o,o}$ term represents the cost incurred by an establishment in subsector s from hiring an employee who remains within their trained occupation. New hires, fresh from occupational training, are treated identically to existing workers for this stage. It should be noted that there are no corner solutions, i.e., $l_s^o(\zeta) > 0$ for all ζ, s, and o.

This stage is called "subsector-occupation autarky" because it is similar to the "autarky" scenario in international trade, when production happens under the constraint that all products must be produced locally. Subsector-Occupation autarky is a scenario where occupational hiring is done "locally" within the occupation space. Furthermore, since the cost of hiring a new worker is the same as the cost of hiring an old worker under the constraint the hiring must be within-subsector, one can interpret the constraint as being that hiring by establishments must not only be within-occupation but also within-subsector as well, i.e., no poaching is allowed. In summary, the "autarky" scenario is one where the labor market operates under the constraint that hiring is done "locally" within the occupation and subsector space.

Stage 2: Cross-Subsector and Cross-Occupation Hiring (Trade)

In the second stage, establishments can hire employees across all subsectors and occupations. In this stage, the problem becomes:

$$\min_{\{l_s^o(\zeta) \mid \forall s, \forall o, \zeta \in \Omega\}} c_s = \left(\sum_o \int_\Omega \overline{c_s^o}(\zeta)\ l_s^o(\zeta)\ d\zeta \right) \tag{38}$$

The output constraint and the labor aggregation defining L_s^o does not change. The cost-of-employment does change. It is now defined by:

$$\overline{c_s^o}(\zeta) = \min_{s', o'} \left\{ c_{s,s'}^{o,o'}(\zeta) \right\}. \tag{39}$$

Thus, the establishment hires the employee with the lowest cost, comparing potential hires across all (o', s') origin points.

Why would employee productivity differ across origins? Following Kortum (1997), if the application of skills is determined by the best available idea, then the resulting productivity distribution is Frechét. Here, the best idea from each ζ type is selected, justifying the use of the Frechét distribution. Workers, regardless of current employment, present ideas for how their skills can be applied in the target (o, s) environment. Worth noting is that these skills can be either existing or hypothetical. They will be hypothetical under the scenario where either $o \neq o'$ or $s \neq s'$. In the latter scenario, workers are aware of the skills they will learn as part of the retraining process associated with their next job, and the ideas they propose will reflect that understanding. Establishments assess these proposals and hire the candidate with the highest expected productivity.

Crucially, employees are evaluated based on their approach to skill application rather than existing formal occupational skills. What is stopping employees from finding out the ideas that managers want to hear and simply repeating them? Moreover, why do managers not simply hire workers who they then instruct on what their approach should be? The answer lies in the nature of the contribution that workers are making. Workers are not helping establishments create new products, but they are still engaged in innovation. Their ideas introduce process innovations that help establishments produce more efficiently, and, since process innovations require ideas that are on the knowledge frontier, it is assumed that neither workers nor managers have access to the full range of possible ideas.

Nevertheless, managers are in a position where they can assess the quality of an idea given to them and accurately judge how productive a worker will be who employs them. The rationale for accurate ex ante estimates by managers but not employees draws inspiration from Hayek's (1988) notion of "dispersed knowledge." It is assumed that the ability to judge the productivity associated with particular ideas requires detailed knowledge about the subsector-specific technological milieu and knowledge environment and that no single person can have that level of understanding for every single subsector. Managers, however, can accurately assess the productivity implications of proposals within their own subsector, even if they themselves cannot enumerate all potential ideas, because their occupational role requires it of them. In the end, the market is one where applicants effectively act as idea entrepreneurs, pitching their strategies for process improvement, without the ability to accurately assess the productivity implications of their ideas, and where managers select the best idea based on accurate knowledge of the productivity associated with each.

Assumptions about Labor Market Clearing Process

After rehiring, inflows and outflows across (s, o) pairs need not balance. The model allows potential imbalances because it assumes that workers will always work when given the opportunity and that representative employees can flexibly adjust hours worked with no upper bound on how many hours are worked. It should be noted that L_s^o represents output, not hours worked or headcount. For a computer scientist, this may mean completed code files; for a tax lawyer, completed tax filings. Higher productivity reduces the cost per output unit. Thus, if a worker of type ζ provides greater output post-matching, it should be interpreted as increased efficiency, not longer working hours per se.

With that being said, not considering an upper bound on the difference between the values of $\sum_s L_s^o$ between the first and second round does allow for a situation where the additional demand for labor output from occupation o could only be met with unrealistic work hours. To resolve this issue,

two things would need to be done. First, complex wage dynamics would need to be introduced that cause wages to rise for occupations that experience much higher demand in the second stage of hiring compared to the first. Second, more than two stages of hiring would need to be incorporated into the model so that higher wages immediately lead to new occupations being trained. The latter would be done with an aim of ensuring a no-arbitrage outcome between occupations after some point. Both of these steps are outside of the focus of this analysis. They would make the model needlessly complex with no added benefit given the research question of this paper. Therefore, it will be assumed that inflows and outflows across (s, o) pairs need not balance, with the stipulation that L_s^o represents output, not hours worked or headcount.

A.2

Below, a closed-form solution for C_s^o is sought. The solution involves using $G_s^o(c)$ as was defined in Subsubsection 3.2.2. Instead of integrating across $\boldsymbol{\zeta}$, the integral can be rewritten to integrate along positive real values where the expression being integrated is now reformulated to be an expression containing c, which is interpreted as a random variable, and that expression is then multiplied by the derivative of the cumulative distribution function of c, i.e., its PDF. This reformulation is treating $\overline{c_s^o}$ as a random variable, which it effectively is before the productivity terms in $\{z_{s,s'}^{o,o'} \mid \forall s' \; \forall o'\}$ become realized. Despite the fact that $\overline{c_s^o}$ is a function of $\boldsymbol{\zeta}$, we know, from Subsubsection 3.2.2, that its distribution does not depend on $\boldsymbol{\zeta}$. This reformulation in terms of random variable c makes it patently clear the the expression inside the integral is integrable and allows for the closed form solution to be easily computed:

$$C_s^o = \Big(\int_{\Omega} \overline{c_s^o}(\boldsymbol{\zeta})^{(1-\eta_s^o)} d\boldsymbol{\zeta} \Big)^{\frac{1}{1-\eta_s^o}} \iff$$

$$(C_s^o)^{1-\eta_s^o} = \int_0^{\infty} c^{(1-\eta_s^o)} dG_s^o(c) \iff$$

$$(C_s^o)^{1-\eta_s^o} = \int_0^{\infty} c^{(1-\eta_s^o)} (\frac{d}{dc}(1 - e^{-\phi_s^o c^{\theta^o}})) dc \iff$$

$$(C_s^o)^{1-\eta_s^o} = \int_0^{\infty} c^{(1-\eta_s^o)} (\phi_s^o \theta^o c^{(\theta^o - 1)} e^{-\phi_s^o c^{\theta^o}}) dc \iff$$

$$(C_s^o)^{1-\eta_s^o} = \phi_s^o \theta^o \int_0^{\infty} c^{(\theta^o - \eta_s^o)} e^{-\phi_s^o c^{\theta^o}} dc \iff$$

At this point, we let $x = c^{\theta^o} \phi_s^o$ and perform a change of variables:

$$(C_s^o)^{1-\eta_s^o} = \int_0^{\infty} (\frac{x}{\phi_s^o})^{\frac{1-\eta_s^o}{\theta^o}} e^{-x} dx \iff$$

$$(C_s^o)^{1-\eta_s^o} = (\phi_s^o)^{-\frac{1-\eta_s^o}{\theta^o}} \int_0^{\infty} (x)^{\frac{1-\eta_s^o}{\theta^o}} e^{-x} dx \iff$$

$$(C_s^o)^{1-\eta_s^o} = (\phi_s^o)^{-\frac{1-\eta_s^o}{\theta^o}} \, \boldsymbol{\Gamma}(\frac{\theta^o + 1 - \eta_s^o}{\theta^o}) \iff$$

$$C_s^o = (\phi_s^o)^{-\frac{1}{\theta^o}} \ \mathbf{\Gamma}(\frac{\theta^o + 1 - \eta_s^o}{\theta^o})^{\frac{1}{1-\eta_s^o}}$$

We know the following:

$$\phi_s^o = \sum_{o'} \sum_{s'} T_s^o (w_{s'}^{o'} d_{s,s'}^{o,o'})^{-\theta^o} = T_s^o \sum_{o'} \sum_{s'} (w_{s'}^{o'} d_{s,s'}^{o,o'})^{-\theta_o}$$

That means that if the gamma-distribution term is defined as M_s^o, then C_s can be expressed as:

$$C_s^o = (T_s^o \sum_{o'} \sum_{s'} (w_{s'}^{o'} d_{s,s'}^{o,o'})^{-\theta_o})^{-\frac{1}{\theta^o}} M_s^o \iff$$

$$C_s^o = (T_s^o)^{-\frac{1}{\theta^o}} \Big(\sum_{o'} \sum_{s'} (w_{s'}^{o'} d_{s,s'}^{o,o'})^{-\theta_o} \Big)^{-\frac{1}{\theta^o}} M_s^o$$

A.3

Let's begin with the following minimization problem:

$$\min_{\{l_s^o(\zeta) \ | \ \forall s \ \forall o \ \zeta \in \Omega\}} c_s = \Big(\sum_o \int_\Omega (\overline{c_s^o}(\zeta) \ l_s^o(\zeta) \ d\zeta) \Big)$$

subject to:

$$\Big(\sum_o \tilde{\beta}_s^o \ (L_s^o)^{\frac{\sigma_s - 1}{\sigma_s}} \Big)^{\frac{\sigma_s}{\sigma_s - 1}} \geq \overline{Q_s}$$

The first order condition with respect to $l_s^o(\zeta)$ is as follows:

$$\frac{d}{d \ l_s^o(\zeta)} \Bigg(\int_\Omega \Big(\overline{c_s^o}(\zeta) \ l_s^o(\zeta) \ d\zeta \Big) - \lambda \Big(\Big(\sum_{o'} \tilde{\beta}_s^{o'} \ (L_s^{o'})^{\frac{\sigma_s - 1}{\sigma_s}} \Big)^{\frac{\sigma_s}{\sigma_s - 1}} - \overline{Q_s} \Big) \Bigg) = 0$$

To simplify the expression, let:

$$B = \sum_{o'} \tilde{\beta}_s^{o'} \ (L_s^{o'})^{\frac{\sigma_s - 1}{\sigma_s}}$$

Also, consider that:

$$\frac{d \ B}{d \ l_s^o(\zeta)} = \tilde{\beta}_s^o \Big(\frac{\sigma_s - 1}{\sigma_s} \Big) (L_s^o)^{\frac{-1}{\sigma_s}} \frac{d \ L_{i,s}^o}{d \ l_{i,s}^o(\zeta)}$$

where:

$$\frac{d \ L_{i,s}^o}{d \ l_{i,s}^o(\zeta)} = \Big(\frac{\eta_{i,s}^o}{\eta_{i,s}^o - 1} \Big) \Bigg(\int_\Omega \Big((l_{i,s}^o(\zeta))^{\frac{(\eta_s^o - 1)}{\eta_s^o}} \ d\zeta \Big)^{\frac{1}{\eta_s^o - 1}} \frac{d}{d \ l_s^o(\zeta)} \Bigg(\int_\Omega \Big((l_s^o(\zeta))^{\frac{(\eta_s^o - 1)}{\eta_s^o}} \ d\zeta \Bigg) \Longrightarrow$$

$$\frac{d\ L^o_{i,s}}{d\ l^o_{i,s}(\zeta)} = \left(\frac{\eta^o_s}{\eta^o_s - 1}\right)\left(\int_\Omega \left((l^o_s(\zeta))^{\frac{(\eta^o_s-1)}{\eta^o_s}}\ d\zeta\right)^{\frac{1}{\eta^o_s-1}} \left(\frac{\eta^o_s - 1}{\eta^o_s}\right)(l^o_s(\zeta))^{-\frac{1}{\eta^o_s}} \implies$$

$$\frac{d\ L^o_{i,s}}{d\ l^o_{i,s}(\zeta)} = \left(\int_\Omega \left((l^o_s(\zeta))^{\frac{(\eta^o_s-1)}{\eta^o_s}}\ d\zeta\right)^{\frac{1}{\eta^o_s-1}} (l^o_s(\zeta))^{-\frac{1}{\eta^o_s}} \implies$$

$$\frac{d\ B}{d\ l^o_s(\zeta)} = \tilde{\beta}^o_s\left(\frac{\sigma_s - 1}{\sigma_s}\right)(L^o_s)^{\frac{-1}{\sigma_s}}\left(\int_\Omega \left((l^o_s(\zeta))^{\frac{(\eta^o_s-1)}{\eta^o_s}}\ d\zeta\right)^{\frac{1}{\eta^o_s-1}} (l^o_s(\zeta))^{-\frac{1}{\eta^o_s}}$$

The first order condition with respect to $l^o_s(\zeta)$, therefore, can be restated as follows:

$$\frac{d}{d\ l^o_s(\zeta)}\left(\int_\Omega \left(\overline{c^o_s}(\zeta)\ l^o_s(\zeta)\ d\zeta\right) - \lambda\left(\left(B\right)^{\frac{\sigma_s}{\sigma_s-1}} - \overline{Q_s}\right)\right) = 0 \implies$$

$$\overline{c^o_s}(\zeta) - \lambda\frac{d}{d\ l^o_s(\zeta)}\left(B\right)^{\frac{\sigma_s}{\sigma_s-1}} = 0 \implies$$

$$\overline{c^o_{i,s}}(\zeta) - \lambda\left(\frac{\sigma_s}{\sigma_s - 1}\right)\left(B\right)^{\frac{1}{\sigma_s-1}}\frac{d\ B}{d\ l^o_s(\zeta)} = 0 \implies$$

$$\overline{c^o_s}(\zeta) - \lambda\left(\frac{\sigma_s}{\sigma_s - 1}\right)\left(B\right)^{\frac{1}{\sigma_s-1}}\tilde{\beta}^o_s\left(\frac{\sigma_s - 1}{\sigma_s}\right)(L^o_s)^{\frac{-1}{\sigma_s}}\left(\int_\Omega \left((l^o_s(\zeta))^{\frac{(\eta^o_s-1)}{\eta^o_s}}\ d\zeta\right)^{\frac{1}{\eta^o_s-1}} (l^o_s(\zeta))^{-\frac{1}{\eta^o_s}} = 0 \implies$$

$$\overline{c^o_s}(\zeta) - \lambda\left(B\right)^{\frac{1}{\sigma_s-1}}\tilde{\beta}^o_s(L^o_s)^{\frac{-1}{\sigma_s}}(L^o_s)^{\frac{1}{\eta^o_s}}(l^o_s(\zeta))^{-\frac{1}{\eta^o_s}} = 0 \implies$$

$$\overline{c^o_s}(\zeta) - \lambda\left(B\right)^{\frac{1}{\sigma_s-1}}\tilde{\beta}^o_s(L^o_s)^{\frac{-\eta^o_s+\sigma_s}{\sigma_s\eta^o_s}}(l^o_s(\zeta))^{-\frac{1}{\eta^o_s}} = 0 \implies$$

$$\overline{c^o_s}(\zeta) = \lambda\left(B\right)^{\frac{1}{\sigma_s-1}}\tilde{\beta}^o_s(L^o_s)^{\frac{-\eta^o_s+\sigma_s}{\sigma_s\eta^o_s}}(l^o_s(\zeta))^{-\frac{1}{\eta^o_s}} \implies$$

$$(\overline{c^o_s}(\zeta))^{1-\eta^o_s} = \left(\lambda\left(B\right)^{\frac{1}{\sigma_s-1}}\tilde{\beta}^o_s(L^o_s)^{\frac{-\eta^o_s+\sigma_s}{\sigma_s\eta^o_s}}\right)^{1-\eta^o_s}(l^o_s(\zeta))^{\frac{\eta^o_s-1}{\eta^o_s}} \implies$$

$$\int_\Omega (\overline{c^o_s}(\zeta))^{1-\eta^o_s}\ d\zeta = \int_\Omega\left(\lambda\left(B\right)^{\frac{1}{\sigma_s-1}}\tilde{\beta}^o_s(L^o_s)^{\frac{-\eta^o_s+\sigma_s}{\sigma_s\eta^o_s}}\right)^{1-\eta^o_s}(l^o_s(\zeta))^{\frac{\eta^o_s-1}{\eta^o_s}}\ d\zeta \implies$$

$$\int_\Omega (\overline{c^o_s}(\zeta))^{1-\eta^o_s}\ d\zeta = \left(\lambda\left(B\right)^{\frac{1}{\sigma_s-1}}\tilde{\beta}^o_s(L^o_s)^{\frac{-\eta^o_s+\sigma_s}{\sigma_s\eta^o_s}}\right)^{1-\eta^o_s}\int_\Omega (l^o_s(\zeta))^{\frac{\eta^o_s-1}{\eta^o_s}}\ d\zeta \implies$$

$$(C_s^o)^{1-\eta_s^o} = \left((\lambda \Big(B\Big)^{\frac{1}{\sigma_s - 1}} \tilde{\beta}_s^o (L_s^o)^{\frac{-\eta_s^o + \sigma_s}{\sigma_s \eta_s^o}} \right)^{1-\eta_s^o} (L_s^o)^{\frac{\eta_s^o - 1}{\eta_s^o}} \implies$$

$$C_s^o = \lambda \Big(B\Big)^{\frac{1}{\sigma_s - 1}} \tilde{\beta}_s^o (L_s^o)^{\frac{-\eta_s^o + \sigma_s}{\sigma_s \eta_s^o}} (L_s^o)^{-\frac{1}{\eta_s^o}} \implies$$

$$C_s^o = \lambda \Big(B\Big)^{\frac{1}{\sigma_s - 1}} \tilde{\beta}_s^o (L_s^o)^{\frac{-\eta_s^o + \sigma_s}{\sigma_s \eta_s^o}} (L_s^o)^{-\frac{\sigma_s}{\eta_s^o \sigma_s}} \implies$$

$$C_s^o = \lambda \Big(B\Big)^{\frac{1}{\sigma_s - 1}} \tilde{\beta}_s^o (L_s^o)^{\frac{-1}{\sigma_s}}$$

The first order condition with respect to λ is as follows:

$$\left(\sum_{o'} \tilde{\beta}_s^{o'} \ (L_s^{o'})^{\frac{\sigma_s - 1}{\sigma_s}} \right)^{\frac{\sigma_s}{\sigma_s - 1}} = \overline{Q_s}$$

Using the definition given to B, the first order condition can be restated as:

$$\Big(B\Big)^{\frac{\sigma_s}{\sigma_s - 1}} = \overline{Q_s} \implies$$

$$\Big(B\Big)^{\frac{1}{\sigma_s - 1}} = \overline{Q_s}^{\frac{1}{\sigma_s}}$$

The expression above can be used in the RHS of the equation with C_s^o from the last first order condition:

$$C_s^o = \lambda \Big(B\Big)^{\frac{1}{\sigma_s - 1}} \tilde{\beta}_s^o (L_s^o)^{\frac{-1}{\sigma_s}} \implies$$

$$C_s^o = \lambda \overline{Q_s}^{\frac{1}{\sigma_s}} \tilde{\beta}_s^o (L_s^o)^{\frac{-1}{\sigma_s}} \implies$$

$$(C_s^o)^{1-\sigma_s} = \lambda^{1-\sigma_s} \overline{Q_s}^{\frac{1-\sigma_s}{\sigma_s}} \tilde{\beta}_s^{o\,1-\sigma s} (L_s^o)^{\frac{\sigma_s - 1}{\sigma_s}} \implies$$

$$(C_s^o)^{1-\sigma_s} = \lambda^{1-\sigma_s} \overline{Q_s}^{\frac{1-\sigma_s}{\sigma_s}} \tilde{\beta}_s^{o\,-\sigma s} \tilde{\beta}_s^o (L_s^o)^{\frac{\sigma_s - 1}{\sigma_s}} \implies$$

$$(C_s^o)^{1-\sigma_s} \tilde{\beta}_s^{o\,\sigma s} = \lambda^{1-\sigma_s} \overline{Q_s}^{\frac{1-\sigma_s}{\sigma_s}} \tilde{\beta}_s^o (L_s^o)^{\frac{\sigma_s - 1}{\sigma_s}} \implies$$

$$\sum_o (C_s^o)^{1-\sigma_s} \tilde{\beta}_s^{o\,\sigma s} = \sum_o \lambda^{1-\sigma_s} \overline{Q_s}^{\frac{1-\sigma_s}{\sigma_s}} \tilde{\beta}_s^o (L_s^o)^{\frac{\sigma_s - 1}{\sigma_s}} \implies$$

$$\sum_o (C_s^o)^{1-\sigma_s} \tilde{\beta}_s^{o\,\sigma s} = \lambda^{1-\sigma_s} \overline{Q_s}^{\frac{1-\sigma_s}{\sigma_s}} \sum_o \tilde{\beta}_s^o (L_s^o)^{\frac{\sigma_s - 1}{\sigma_s}} \implies$$

$$C_s^{1-\sigma_s} = \lambda^{1-\sigma_s}\overline{Q_s}^{\frac{1-\sigma_s}{\sigma_s}}\overline{Q_s}^{\frac{\sigma_s-1}{\sigma_s}} \implies$$

$$C_s^{1-\sigma_s} = \lambda^{1-\sigma_s} \implies$$

$$C_s = \lambda$$

Recall that:

$$C_s^o = \lambda\overline{Q_s}^{\frac{1}{\sigma_s}}\tilde{\beta}_s^o(L_s^o)^{\frac{-1}{\sigma_s}}$$

Now, λ can be replaced with C_s:

$$C_s^o = \lambda\overline{Q_s}^{\frac{1}{\sigma_s}}\tilde{\beta}_s^o(L_s^o)^{\frac{-1}{\sigma_s}} \implies$$

$$C_s^o = C_s\overline{Q_s}^{\frac{1}{\sigma_s}}\tilde{\beta}_s^o(L_s^o)^{\frac{-1}{\sigma_s}}$$

The equation above can be rearranged so that L_s^o is expressed as a function of the other terms:

$$C_s^o = C_s\overline{Q_s}^{\frac{1}{\sigma_s}}\tilde{\beta}_s^o(L_s^o)^{\frac{-1}{\sigma_s}} \implies$$

$$(L_s^o)^{\frac{1}{\sigma_s}} = (C_s/C_s^o)\overline{Q_s}^{\frac{1}{\sigma_s}}\tilde{\beta}_s^o \implies$$

$$L_s^o = (C_s^o/C_s)^{-\sigma_s}\overline{Q_s}\tilde{\beta}_s^{o\sigma_s}$$

Therefore, the labor demand condition is as follows:

$$L_s^o = \left(\frac{C_s^o}{\tilde{\beta}_s^o C_s}\right)^{-\sigma_s}\overline{Q_s}$$

A.4

From Subsection A.3, the following is known:

$$L_s^o = \left(\frac{C_s^o}{\tilde{\beta}_s^o C_s}\right)^{-\sigma_s}\overline{Q_s}$$

Therefore:

$$\frac{L_s^o}{L_s^{o'}} > \frac{L_{s'}^o}{L_{s'}^{o'}} \implies$$

$$\left(\frac{\left(\frac{C_s^o}{\tilde{\beta}_s^o C_s}\right)^{-\sigma_s}\overline{Q_s}}{\left(\frac{C_s^{o\prime}}{\tilde{\beta}_{s'}^{o\prime} C_s}\right)^{-\sigma_s}\overline{Q_s}}\right) > \left(\frac{\left(\frac{C_{s'}^o}{\tilde{\beta}_{s'}^o C_{s'}}\right)^{-\sigma_{s'}}\overline{Q_{s'}}}{\left(\frac{C_{s'}^{o\prime}}{\tilde{\beta}_{s'}^{o\prime} C_{s'}}\right)^{-\sigma_{s'}}\overline{Q_{s'}}}\right)$$

Now, the sufficient conditions are applied and the inequality above is simplified:

$$\left(\frac{(C_s^o)^{-\sigma_s}}{(C_s^{o\prime})^{-\sigma_s}}\right) > \left(\frac{(C_{s'}^o)^{-\sigma_{s'}}}{(C_{s'}^{o\prime})^{-\sigma_{s'}}}\right) \implies$$

$$\frac{C_s^o}{C_s^{o\prime}} < \frac{C_{s'}^o}{C_{s'}^{o\prime}}$$

With the latter holding since $\sigma_s = \sigma_{s'}$. Now, occupation-subsector autarky is assumed:

$$\frac{\widetilde{C_s^o}}{\widetilde{C_s^{o\prime}}} < \frac{\widetilde{C_{s'}^o}}{\widetilde{C_{s'}^{o\prime}}}$$

For the next part, the following notation is used:

$$M_s = \left(\Gamma\left(\frac{\theta^s + 1 - \eta_s}{\theta^s}\right)\right)^{\frac{1}{1-\eta_s}}$$

Using equation (23) from Subsubsection 3.2.2, it follows that:

$$\frac{\widetilde{C_s^o}}{\widetilde{C_s^{o\prime}}} < \frac{\widetilde{C_{s'}^o}}{\widetilde{C_{s'}^{o\prime}}} \implies$$

$$\frac{\left(\widetilde{\phi_s^o}\right)^{-\frac{1}{\theta^s}} M_s}{\left(\widetilde{\phi_s^{o\prime}}\right)^{-\frac{1}{\theta^s}} M_s} < \frac{\left(\widetilde{\phi_{s'}^o}\right)^{-\frac{1}{\theta^{s'}}} M_{s'}}{\left(\widetilde{\phi_{s'}^{o\prime}}\right)^{-\frac{1}{\theta^{s'}}} M_{s'}} \implies$$

$$\frac{\left(\widetilde{\phi_s^o}\right)^{-\frac{1}{\theta^s}}}{\left(\widetilde{\phi_s^{o\prime}}\right)^{-\frac{1}{\theta^s}}} < \frac{\left(\widetilde{\phi_{s'}^o}\right)^{-\frac{1}{\theta^{s'}}}}{\left(\widetilde{\phi_{s'}^{o\prime}}\right)^{-\frac{1}{\theta^{s'}}}}$$

Using equation (22) from Subsubsection 3.2.2, it follows that the inequality above becomes:

$$\frac{(T_s^o)^{-\frac{1}{\theta_s}} w_s^o}{(T_s^{o\prime})^{-\frac{1}{\theta_s}} w_s^{o\prime}} < \frac{(T_{s'}^o)^{-\frac{1}{\theta_{s'}}} w_{s'}^o}{(T_{s'}^{o\prime})^{-\frac{1}{\theta_{s'}}} w_{s'}^{o\prime}} \implies$$

$$\frac{\left(\frac{T_s^o}{(w_s^o)^{\theta_s})}\right)^{-\frac{1}{\theta_s}}}{\left(\frac{T_s^{o'}}{(w_s^{o'})^{\theta_s})}\right)^{-\frac{1}{\theta_s}}} > \frac{\left(\frac{T_{s'}^o}{(w_{s'}^o)^{\theta_{s'}})}\right)^{-\frac{1}{\theta_{s'}}}}{\left(\frac{T_{s'}^{o'}}{(w_{s'}^{o'})^{\theta_{s'}})}\right)^{-\frac{1}{\theta_{s'}}}}$$

A.5

The proof begins with the revealed comparative advantage index of occupation o in subsector s being greater than the revealed comparative advantage index value of occupation o' in subsector s:

$$\left(\frac{L_s^o}{L_s}\right)/\left(\frac{L^o}{L}\right) > \left(\frac{L_s^{o'}}{L_s}\right)/\left(\frac{L^{o'}}{L}\right) \implies$$

$$\left(\frac{L_s^o}{L_o}\right)/\left(\frac{L^s}{L}\right) > \left(\frac{L_s^{o'}}{L_{o'}}\right)/\left(\frac{L^s}{L}\right) \implies$$

$$\left(\frac{L_s^o}{L_o}\right) > \left(\frac{L_s^{o'}}{L_{o'}}\right)$$

At this point, the results derived in Subsection A.3 can be used to replace the labor terms with the cost-of-employment indices. Note that L^o, $L^{o'}$, and L can all be derived by taking sums of the RHS of the labor demand condition derived in Subsection A.3. After replacing each of the labor terms, the following relationship results:

$$\left(\frac{L_s^o}{L_o}\right) > \left(\frac{L_s^{o'}}{L_{o'}}\right) \implies$$

$$\left(\frac{\left(\frac{C_s^o}{\tilde{\beta}_s^o C_s}\right)^{-\sigma_s}\overline{Q_s}}{\sum_{s'}\left(\frac{C_{s'}^o}{\tilde{\beta}_{s'}^o C_{s'}}\right)^{-\sigma_{s'}}\overline{Q_{s'}}}\right) > \left(\frac{\left(\frac{C_s^{o'}}{\tilde{\beta}_s^{o'} C_s}\right)^{-\sigma_s}\overline{Q_s}}{\sum_{s'}\left(\frac{C_{s'}^{o'}}{\tilde{\beta}_{s'}^{o'} C_{s'}}\right)^{-\sigma_{s'}}\overline{Q_{s'}}}\right) \implies$$

$$\left(\frac{\tilde{\beta}_s^{o\,\sigma_s}\left(C_s^o\right)^{-\sigma_s}}{\sum_{s'}\left(\frac{C_{s'}^o}{\tilde{\beta}_{s'}^o C_{s'}}\right)^{-\sigma_{s'}}\overline{Q_{s'}}}\right) > \left(\frac{\tilde{\beta}_s^{o'\,\sigma_s}\left(C_s^{o'}\right)^{-\sigma_s}}{\sum_{s'}\left(\frac{C_{s'}^{o'}}{\tilde{\beta}_{s'}^{o'} C_{s'}}\right)^{-\sigma_{s'}}\overline{Q_{s'}}}\right) \implies$$

$$\left(\frac{\left(C_s^o\right)^{-\sigma_s}}{\sum_{s'}\left(\frac{C_{s'}^o}{\tilde{\beta}_{s'}^o C_{s'}}\right)^{-\sigma_{s'}}\overline{Q_{s'}}}\right) > \left(\frac{\tilde{\beta}_s^{o'}}{\tilde{\beta}_s^o}\right)^{\sigma_s}\left(\frac{\left(C_s^{o'}\right)^{-\sigma_s}}{\sum_{s'}\left(\frac{C_{s'}^{o'}}{\tilde{\beta}_{s'}^{o'} C_{s'}}\right)^{-\sigma_{s'}}\overline{Q_{s'}}}\right) \implies$$

$$\left(\frac{C_s^o}{C_s^{o'}}\right)^{-\sigma_s} > \left(\frac{\tilde{\beta}_s^{o'}}{\tilde{\beta}_s^o}\right)^{\sigma_s} \left(\frac{\sum_{s'}\left(\frac{C_{s'}^o}{\tilde{\beta}_{s'}^o C_{s'}}\right)^{-\sigma_{s'}} \overline{Q_{s'}}}{\sum_{s'}\left(\frac{C_{s'}^{o'}}{\tilde{\beta}_{s'}^{o'} C_{s'}}\right)^{-\sigma_{s'}} \overline{Q_{s'}}}\right)$$

At this point, the condition $\tilde{\beta}_{s'}^{o'} = 1 \; \forall s, \forall o$ gets applied, which leads to:

$$\left(\frac{C_s^o}{C_s^{o'}}\right)^{-\sigma_s} > \left(\frac{\sum_{s'}\left(\frac{C_{s'}^o}{C_{s'}}\right)^{-\sigma_{s'}} \overline{Q_{s'}}}{\sum_{s'}\left(\frac{C_{s'}^{o'}}{C_{s'}}\right)^{-\sigma_{s'}} \overline{Q_{s'}}}\right)$$

Consider that the above can be rewritten in terms of the absolute cost disadvantage:

$$\left(\frac{C_s^o}{C_s^{o'}}\right)^{-\sigma_s} > \left(\frac{\sum_{s'}\left(\frac{C_{s'}^o}{C_{s'}}\right)^{-\sigma_{s'}} \overline{Q_{s'}}}{\sum_{s'}\left(\frac{C_{s'}^{o'}}{C_{s'}}\right)^{-\sigma_{s'}} \overline{Q_{s'}}}\right) \implies$$

$$\left(\frac{C_s^o}{C_{o'}}\right)^{-\sigma_s} > \frac{(ACD_s^o)^{-1}}{(ACD_{o'})^{-1}} \implies$$

$$\left(\frac{C_s^o}{C_s^{o'}}\right)^{\sigma_s} < \frac{(ACD_o)}{(ACD_{o'})} \implies$$

$$\frac{C_s^o}{(ACD_o)^{\frac{1}{\sigma_s}}} < \frac{C_s^{o'}}{(ACD_{o'})^{\frac{1}{\sigma_s}}}$$

Now, let's assume subsector-occupation autarky. Then, the following holds:

$$\frac{\widetilde{C_s^o}}{(\widetilde{ACD_o})^{\frac{1}{\sigma_s}}} < \frac{\widetilde{C_s^{o'}}}{(\widetilde{ACD_{o'}})^{\frac{1}{\sigma_s}}} \implies$$

$$\frac{\left(\widetilde{\phi_s^o}\right)^{-\frac{1}{\theta^s}} M_s}{\left(\sum_{s'}\left(\frac{\left(\widetilde{\phi_{s'}^o}\right)^{-\frac{1}{\theta^{s'}}} M_{s'}}{\widetilde{C_{s'}}}\right)^{-\sigma_{s'}} \overline{Q_{s'}}\right)^{\frac{1}{\sigma_s}}} < \frac{\left(\widetilde{\phi_s^{o'}}\right)^{-\frac{1}{\theta^s}} M_s}{\left(\sum_{s'}\left(\frac{\left(\widetilde{\phi_{s'}^{o'}}\right)^{-\frac{1}{\theta^{s'}}} M_{s'}}{\widetilde{C_{s'}}}\right)^{-\sigma_{s'}} \overline{Q_{s'}}\right)^{\frac{1}{\sigma_s}}} \implies$$

$$\frac{(T_s^o)^{-\frac{1}{\theta_s}} w_s^o}{\left(\sum_{s'}\left(\frac{(T_{s'}^o)^{-\frac{1}{\theta_{s'}}} w_s^o M_{s'}}{\widetilde{C_{s'}}}\right)^{-\sigma_{s'}} \overline{Q_{s'}}\right)^{\frac{1}{\sigma_s}}} < \frac{(T_s^{o'})^{-\frac{1}{\theta_s}} w_s^{o'}}{\left(\sum_{s'}\left(\frac{(T_{s'}^{o'})^{-\frac{1}{\theta_{s'}}} w_s^{o'} M_{s'}}{\widetilde{C_{s'}}}\right)^{-\sigma_{s'}} \overline{Q_{s'}}\right)^{\frac{1}{\sigma_s}}}$$

To simplify the expression above, once again, ρ_s^o is used:

$$\rho_s^o = \frac{T_s^o}{(w_s^o)^{\theta_s}}$$

Note that the expression above becomes:

$$\frac{(\rho_s^o)^{-\frac{1}{\theta_s}}}{\left(\sum_{s'}\left(\frac{(\rho_{s'}^o)^{-\frac{1}{\theta_{s'}}} M_{s'}}{\widetilde{C_{s'}}}\right)^{-\sigma_{s'}} \overline{Q_{s'}}\right)^{\frac{1}{\sigma_s}}} < \frac{(\rho_s^{o'})^{-\frac{1}{\theta_s}}}{\left(\sum_{s'}\left(\frac{(\rho_{s'}^{o'})^{-\frac{1}{\theta_{s'}}} M_{s'}}{\widetilde{C_{s'}}}\right)^{-\sigma_{s'}} \overline{Q_{s'}}\right)^{\frac{1}{\sigma_s}}} \implies$$

$$\frac{(\rho_s^o)^{\frac{1}{\theta_s}}}{\left(\sum_{s'}\left(\frac{(\rho_{s'}^o)^{-\frac{1}{\theta_{s'}}} M_{s'}}{\widetilde{C_{s'}}}\right)^{-\sigma_{s'}} \overline{Q_{s'}}\right)^{\frac{-1}{\sigma_s}}} > \frac{(\rho_{s'}^o)^{\frac{1}{\theta_{s'}}}}{\left(\sum_{s'}\left(\frac{(\rho_{s'}^{o'})^{-\frac{1}{\theta_{s'}}} M_{s'}}{\widetilde{C_{s'}}}\right)^{-\sigma_{s'}} \overline{Q_{s'}}\right)^{\frac{-1}{\sigma_s}}}$$

Now, recall that:

$$\widetilde{C_{s'}} = \left(\sum_{o''}(\widetilde{C_{s'}^{o''}})^{1-\sigma_s}\right)^{\frac{1}{1-\sigma_s}} = \left(\sum_{o''}\left((\widetilde{\phi_s^o})^{-\frac{1}{\theta^s}} M_s\right)^{1-\sigma_s}\right)^{\frac{1}{1-\sigma_s}} \implies$$

$$\widetilde{C_{s'}} = M_{s'}\left(\sum_{o''}\left((T_{s'}^{o''})^{-\frac{1}{\theta_{s'}}} w_s^{o''}\right)^{1-\sigma_{s'}}\right)^{\frac{1}{1-\sigma_{s'}}} \implies$$

$$\widetilde{C_{s'}} = M_{s'}\left(\sum_{o''}\left(\left(\frac{T_{s'}^{o''}}{(w_s^{o''})^{\theta_{s'}}}\right)^{-\frac{1}{\theta_{s'}}}\right)^{1-\sigma_{s'}}\right)^{\frac{1}{1-\sigma_{s'}}} \implies$$

$$\widetilde{C_{s'}} = M_{s'}\left(\sum_{o''}(\rho_{s'}^{o''})^{\frac{\sigma_{s'}-1}{\theta_{s'}}}\right)^{\frac{1}{1-\sigma_{s'}}}$$

With that in mind, consider that:

$$\frac{(\rho_s^o)^{\frac{1}{\theta_s}}}{\left(\sum_{s'}\left(\frac{(\rho_{s'}^o)^{-\frac{1}{\theta_{s'}}} M_{s'}}{\widetilde{C_{s'}}}\right)^{-\sigma_{s'}} \overline{Q_{s'}}\right)^{\frac{-1}{\sigma_s}}} > \frac{(\rho_s^{o'})^{\frac{1}{\theta_s}}}{\left(\sum_{s'}\left(\frac{(\rho_{s'}^{o'})^{-\frac{1}{\theta_{s'}}} M_{s'}}{\widetilde{C_{s'}}}\right)^{-\sigma_{s'}} \overline{Q_{s'}}\right)^{\frac{-1}{\sigma_s}}} \implies$$

$$\frac{(\rho_s^o)^{\frac{1}{\theta_s}}}{(\sum_{s'}\left(\frac{(\rho_{s'}^o)^{-\frac{1}{\theta_{s'}}} M_{s'}}{M_{s'}\left(\sum_{o''}(\rho_{s'}^{o''})^{\frac{\sigma_{s'}-1}{\theta_{s'}}}\right)^{\frac{1}{1-\sigma_{s'}}}}\right)^{-\sigma_{s'}} \overline{Q_{s'}})^{\frac{-1}{\sigma_s}}} > \frac{(\rho_s^{o'})^{\frac{1}{\theta_s}}}{(\sum_{s'}\left(\frac{(\rho_{s'}^{o'})^{-\frac{1}{\theta_{s'}}} M_{s'}}{M_{s'}\left(\sum_{o''}(\rho_{s'}^{o''})^{\frac{\sigma_{s'}-1}{\theta_{s'}}}\right)^{\frac{1}{1-\sigma_{s'}}}}\right)^{-\sigma_{s'}} \overline{Q_{s'}})^{\frac{-1}{\sigma_s}}} \implies$$

$$\frac{(\rho_s^o)^{\frac{1}{\theta_s}}}{(\sum_{s'}\left(\frac{(\rho_{s'}^o)^{-\frac{1}{\theta_{s'}}}}{\left(\sum_{o''}((\rho_{s'}^{o''})^{-\frac{1}{\theta_{s'}}})^{(1-\sigma_{s'})}\right)^{\frac{1}{1-\sigma_{s'}}}}\right)^{-\sigma_{s'}}\overline{Q_{s'}})^{\frac{-1}{\sigma_s}}} > \frac{(\rho_{s'}^o)^{\frac{1}{\theta_{s'}}}}{(\sum_{s'}\left(\frac{(\rho_{s'}^{o'})^{-\frac{1}{\theta_{s'}}}}{\left(\sum_{o''}((\rho_{s'}^{o''})^{-\frac{1}{\theta_{s'}}})^{(1-\sigma_{s'})}\right)^{\frac{1}{1-\sigma_{s'}}}}\right)^{-\sigma_{s'}}\overline{Q_{s'}})^{\frac{-1}{\sigma_s}}}$$

Recall the following indices:

$$APD_s = \left(\sum_{o''}((\rho_{s'}^{o''})^{-\frac{1}{\theta_{s'}}})^{(1-\sigma_{s'})}\right)^{\frac{1}{1-\sigma_{s'}}}$$

$$APD^o = \left(\sum_{s'}\left(\frac{(\rho_s^o)^{-\frac{1}{\theta_s}}}{APD_s}\right)^{-\sigma_s}\overline{Q_s}\right)^{-1}$$

Writing the inequality in terms of the indices above leads to:

$$\frac{(\rho_s^o)^{\frac{1}{\theta_s}}}{(\sum_{s'}\left(\frac{(\rho_{s'}^o)^{-\frac{1}{\theta_{s'}}}}{\left(\sum_{o''}((\rho_{s'}^{o''})^{-\frac{1}{\theta_{s'}}})^{(1-\sigma_{s'})}\right)^{\frac{1}{1-\sigma_{s'}}}}\right)^{-\sigma_{s'}}\overline{Q_{s'}})^{\frac{-1}{\sigma_s}}} > \frac{(\rho_{s'}^o)^{\frac{1}{\theta_{s'}}}}{(\sum_{s'}\left(\frac{(\rho_{s'}^{o'})^{-\frac{1}{\theta_{s'}}}}{\left(\sum_{o''}((\rho_{s'}^{o''})^{-\frac{1}{\theta_{s'}}})^{(1-\sigma_{s'})}\right)^{\frac{1}{1-\sigma_{s'}}}}\right)^{-\sigma_{s'}}\overline{Q_{s'}})^{\frac{-1}{\sigma_s}}} \implies$$

$$\frac{(\rho_s^o)^{\frac{1}{\theta_s}}}{(APD^o)^{-\frac{1}{\sigma_s}}} > \frac{(\rho_s^o)^{\frac{1}{\theta_s}}}{(APD^{o'})^{-\frac{1}{\sigma_s}}}$$

Thus, the proof is complete.

A.6 Relationship to Jara-Figueroa et al. (2018)

The regression used in this paper has some similarities to the one used in Jara-Figueroa et al. (2018), but some differences as well. The crucial difference is that the regression specification in this paper is designed with the goal of finding whether complementarity has an effect on an establishment's productivity. Jara-Figueroa et al. (2018), on the other hand, use the notion of complementarity, including equation (8), to provide an alternative method of reconstructing industry-knowledge and occupational-knowledge scores. Their goal is to find effect of industry and occupational knowledge on a firm's size.

In the model, I showed how the component of occupational-knowledge and industry-knowledge can be formalized into the Eaton and Kortum framework. The $d_{s,s'}^{o,o'}$ variable is able to capture the benefits associated with occupation and subsector switched for which there is a high degree of occupational or subsector-related knowledge. I made clear that if $d_{s,s'}^{o,o'} = 1$ for all occupation and subsector pairs, the complementarity score becomes a better measure of the technology parameter. Therefore, if occupation and industry knowledge play a large role in the ease in which people

can switch occupations or subsectors, the complementarity score becomes a less useful measure. Jara-Figueroa et al. (2018) argue for the important role of occupational and industry-knowledge, but it should be noted that their regression did not attempt to measure the possibility of complementarity between subsector-occupation pairs. The effect of such complementarity may instead have been captured by their occupation and industry knowledge scores.

To make this distinction intuitively clear, one should consider that the industry-occupation complementarity score depends on the set of roles within an establishment. The knowledge scores are a measure of the knowledge that employees bring into the establishment, but how they use that knowledge, will depend on the roles that the establishment makes available for them. For example, suppose a startup is made up of employees who all have experience in managing and programming. The startup can decide to put 80 percent of them in programming roles and 20 percent in managerial roles. Alternatively, it can decide to put 98 percent of them in programming roles and two percent of them in managerial roles. The occupational knowledge brought in is the same in both scenarios, but the roles the employees exercise within the firm are different.

To summarize, the industry-occupation complementarity scores depend on the occupational groups hired, which, in effect, makes it a score that depends on the set of occupational roles an establishment chooses to create for itself. Knowledge scores alone do not capture the ability occupational groups have in influencing production because the power with which each occupational group has to influence production varies by industry, and it depends on position rather than only on knowledge.